

智能系统中的协同进化：迈向超越人类设计的自主进化

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摘要

智能体系统在部署后持续改进的期望日益增长，然而单一实体的自我进化往往受限于静态学习情境，例如固定的任务与反馈。本综述聚焦于智能系统中的协同进化，这是一种多组件的自我进化形式，其中多个智能体及其环境相互施加适应性压力。为梳理现有文献，本文提出一个渐进式三阶段分类体系，追踪系统如何逐步摆脱人类工程约束。智能体间协同进化研究智能体如何通过动态同伴进行适应，包括对抗性、协作性与组织性适应。智能体与环境协同进化将这一循环扩展至随智能体变化的自适应任务、反馈与交互空间。元协同进化进一步探索使进化机制本身可进化的可能性。本文还讨论了评估此类系统、跨多组件扩展以及确保日益自主的进化过程安全可控等开放挑战。本综述为构建稳健且开放式的智能体系统提供了统一基础，使其能够超越固定的人类设计路径持续改进。

1 引言

人工智能的最新进展已将关注点从孤立模型转向智能体系统，后者能够自主地与其他智能体及外部环境交互（Guo 等，2024；Plaat 等，2025；Luo 等，2025）。此类系统可通过外部框架增强（Ning 等，2026；Yao 等，2026b），包括工具（Xu 等，2025b）、记忆（Du，2026）与技能（Zhou 等，2026）。随着智能体系统日益复杂，一个核心问题是如何在部署后持续改进（Gao 等，2026）。自我进化是智能体系统持续改进的主要范式，其中

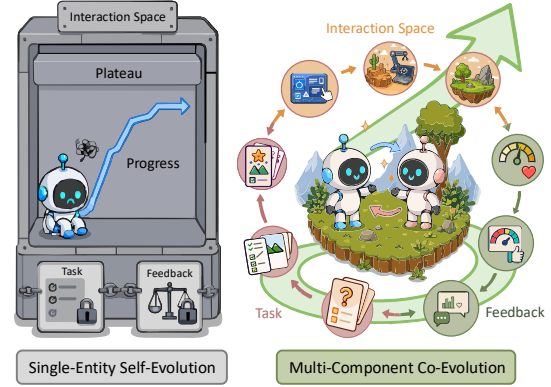


图 1：单一实体自我进化与多组件协同进化的比较。

智能体在没有人类干预的情况下，持续从经验、反馈和失败中更新自身。一种重要形式是单实体自我进化，其中更新发生在一个智能体内部，例如改进其模型骨干（Zhai 等人，2025a；Wu 等人，2025）、更新记忆（Suzgun 等人，2026）或提升技能（Xiao 等人，2026）。然而，它可能仍受固定外部条件的限制，如图 1 所示。这一局限性呼应了红皇后效应（Van Valen，2014）：持续进步需要相互适应，而非仅单方面的适应。协同进化通过使对方具有适应性来扩展自我进化（Hillis，1990）。在智能体系统中，智能体可能与对手或协作者等适应性同伴共同进化（Pan 等人，2026b），或与任务、反馈和交互空间等环境共同进化（Guo 等人，2025；Liu 等人，2026h）。近期工作进一步提出了一种元层面的扩展，其中进化机制本身变得可进化（Iacob 等人，2026）。在这些情境下，多个组件共同适应并持续重塑彼此的进一步进化，而不仅仅是交换信息或进行交互。尽管关注度日益增长，但尚无综述以

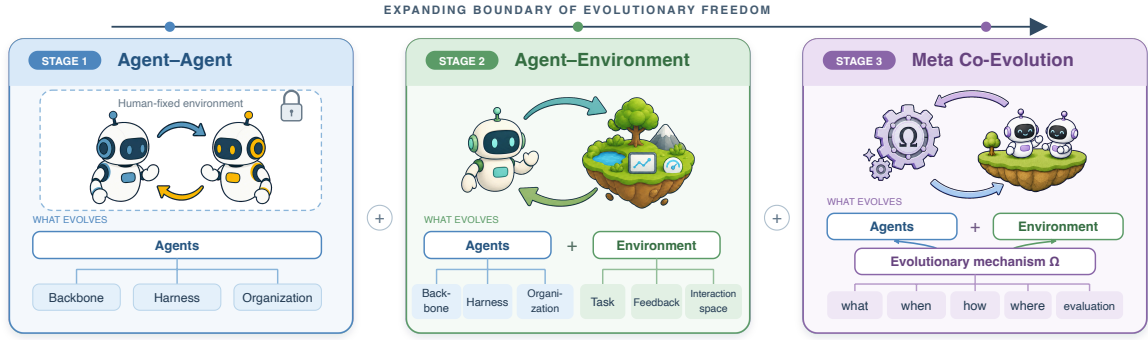


图 2: 协同进化的渐进式分类法, 反映了进化自由度的不断扩展边界。

此类协同进化智能体系统为中心。它们考察了基于大语言模型的智能体和多智能体系统（Guo 等人, 2024; Plaatt 等人, 2025; Luo 等人, 2025）、智能体组件与框架工程（Xu 等人, 2025b; Du, 2026; Zhou 等人, 2026; Meng 等人, 2026），以及面向交互式智能体的环境扩展（Huang 等人, 2025c）。关于自我进化智能体的综述（Gao 等人, 2026; Xiang 等人, 2026）与我们的范围最为接近, 但将协同进化视为一个子主题, 而非核心组织轴。我们通过提出以下问题来填补这一空白: 哪些组件在协同进化, 进化自由度的边界如何跨组件扩展, 以及这一过程的最终形态可能是什么。为了组织这些文献, 我们提出了一种渐进式三阶段分类法, 该分类法以系统被允许进化的范围不断扩展为结构, 追踪系统如何逐步摆脱人类设计的约束。(1) 智能体-智能体协同进化捕捉了在固定环境中进化同伴之间的相互适应, 包括其模型骨干、框架组件和组织结构的变化。

(2) 智能体-环境协同演化将适应性扩展至环境组件, 包括任务、反馈和交互空间。(3) 元协同演化捕捉了一个新兴阶段, 其中演化机制本身变得具有适应性, 使系统能够修订演化什么、何时演化、如何演化、在哪里演化, 以及如何评估。这一进展将演化自由度从仅智能体扩展到智能体与环境, 最终扩展到支配其演化的规则, 为开放性提供了一条可能的路径（Stanley 等, 2017; Hughes 等, 2024），其中系统持续改进而非收敛到固定终点。本综述做出三项贡献。(1) 据我们所知, 我们提供了首个聚焦的综述

关于智能体系统中的协同演化, 区分了相互适应与单纯的交互或信息交换。(2) 我们通过一个渐进的三阶段分类法组织文献, 追踪演化自由度如何从仅智能体扩展到智能体与环境, 最终扩展到可演化的演化机制。(3) 我们讨论了评估、扩展和治理日益自主的协同演化系统的开放挑战和未来方向。本文其余部分组织如下。第 2 节定义形式基础并介绍我们的三阶段渐进分类法。第 3–5 节回顾三个阶段的方法。第 6 节讨论挑战和未来方向。第 7 节总结。

2 预备知识

2.1 什么是智能体系统?

智能体系统 $S = (A, E)$ 由智能体集体 A 和环境 E 组成。智能体是一个系统单元, 能够自主地与其他智能体或环境交互。为了追踪演化, 我们将每个智能体视为一个统一的演化单元, 并用 $a_i = (m_i, h_i)$ 表示: 模型骨干 m_i 和装备 h_i （Yao 等, 2026b），例如记忆、工具、技能、提示或工作流。在实践中, h_i 可能仅包含其中一些组件, 或者不包含任何组件。智能体可能共享相同的骨干, 但如果它们的装备实例化了不同的角色或目标, 则仍然保持不同。当任一组件发生变化时, 智能体发生演化:

$$\Delta a_i \neq 0 \iff (\Delta m_i \neq 0) \vee (\Delta h_i \neq 0).$$

智能体并非无序集合。它们由 Π 结构化, 该结构编码了它们的角色、通信拓扑和分工:

$$A = (\{a_1, \dots, a_n\}, \Pi).$$

A evolves whenever an a_i or Π changes.

环境 E 是智能体所作用并从中接收反馈的 A 之外的一切。轨迹 τ^t 记录了截至步骤 t 系统累积的思考—行动—观察循环序列 (Yao 等人, 2023)。

2.2 什么是共同进化？

设 Ω 为驱动状态转换的进化机制，写作 $S^{t+1} = \Omega(S^t, \tau^t)$ 。它规定了什么可以进化、何时触发进化、变体如何生成或更新、进化发生在何处，以及如何评估进化质量。在本综述中，共同进化是自进化的一种耦合形式，其中至少两个进化单元联合适应并持续重塑彼此的进一步进化，而不仅仅是交换信息或交互。它要求：

$$\begin{aligned} S^{t+1} &= \Omega(S^t, \tau^t), \quad \exists x \neq y \in S^t : \\ x^{t+1} &\neq x^t, \quad y^{t+1} \neq y^t, \\ &\text{进化压力} \\ x &\longleftrightarrow y. \end{aligned}$$

2.3 三阶段分类法

我们的三阶段分类法（图 2）遵循系统被允许进化范围的扩展。从一个阶段到下一个阶段，人为设计的约束被逐步移除。

阶段 1——智能体—智能体共同进化。当智能体不再针对静态对手学习时，共同进化便开始了。随着智能体相互响应，每个智能体的进展改变了其同伴面临的挑战和机遇，集体也可能重组其结构 Π 。这在智能体集体内部创造了一个耦合的进化过程：

$$A^{t+1} = (\{a_i^{t+1}\}_{i=1}^n, \Pi^{t+1}) = \Omega(A^t, E, \tau^t).$$

第二阶段——智能体与环境的协同演化。这种耦合过程能够持续改变智能体，但无法改变它们交互的环境。当环境保持固定时，智能体所能遇到的新经验和新压力就会受到限制。因此，第二阶段将这一过程扩展到环境本身，允许智能体的行为重塑整体条件，而这些条件随后又会反过来塑造智能体：

$$(A^{t+1}, E^{t+1}) = \Omega(A^t, E^t, \tau^t).$$

第三阶段——元协同演化。在第二阶段中，演化机制仍然由人类设计，因此系统可以改变其内容，但无法改变它们

变化的方式。因此，我们将元协同演化定义为这样一个阶段：低层协同演化系统通过自生成的修订过程 Γ^t 进一步修改其演化机制：

$$\begin{aligned} \Omega^{t+1} &= \Gamma^t(S^t, \Omega^t, \tau^t), \\ S^{t+1} &= \Omega^{t+1}(S^t, \tau^t). \end{aligned}$$

This recursion provides a pathway toward open-endedness (Stanley et al., 2017; Hughes et al., 2024), where the system continues to generate meaningful novelty rather than converging to a fixed endpoint, characterized by continuous novelty, $\Omega^{t+1} \neq \Omega^t$, and unbounded divergence, with adaptive capability $\mathcal{H}$ satisfying $\lim_{t \rightarrow \infty} \mathcal{H}(S^t, \Omega^t) = \infty$.

3 智能体与智能体的协同演化

这一阶段研究智能体集体内部的协同演化 $A = (\{a_i\}_{i=1}^n, \Pi)$ ，其中每个智能体改进的主要来源是周围其他正在演化的智能体。我们将这一阶段组织为三种模式：根据智能体目标关系区分的对抗性协同演化和协作性协同演化，以及进一步调整智能体角色和交互结构的演化型智能体组织。

3.1 对抗性智能体

对抗性智能体通过相互对立的目标共同进化，每一方都因击败对方而获得奖励，因此一个智能体的进步会直接提高其对手面临的难度。本文根据压力扩展的方式组织相关文献，从两个进化方之间的成对压力到多个对抗性智能体之间的多源压力。

3.1.1 成对对抗压力 这种共同进化出现在两个具有不同角色的对抗性智能体之间，可以追溯到生成对抗网络 (GAN) (Goodfellow 等人, 2014)，其中生成器生成假图像，判别器学习将其与真实图像区分开来，两者相互促进。同样的思想很快被用于具身控制。鲁棒对抗强化学习 (RARL) (Pinto 等人, 2017) 训练控制策略对抗一个添加不稳定力的对手，使其对真实扰动具有鲁棒性，而不是对固定噪声模式。通过在格斗游戏 (Bansal 等人, 2018) 和捉迷藏 (Baker 等人, 2020) 中的竞争性自我博弈，智能体在相互升级的过程中学会了工具使用和反制策略。这一思想后来扩展到语言领域，其中自我博弈对抗生成 (SPAG) (Cheng 等人, 2024) 训练词语猜测游戏中的两个智能体，其中一个

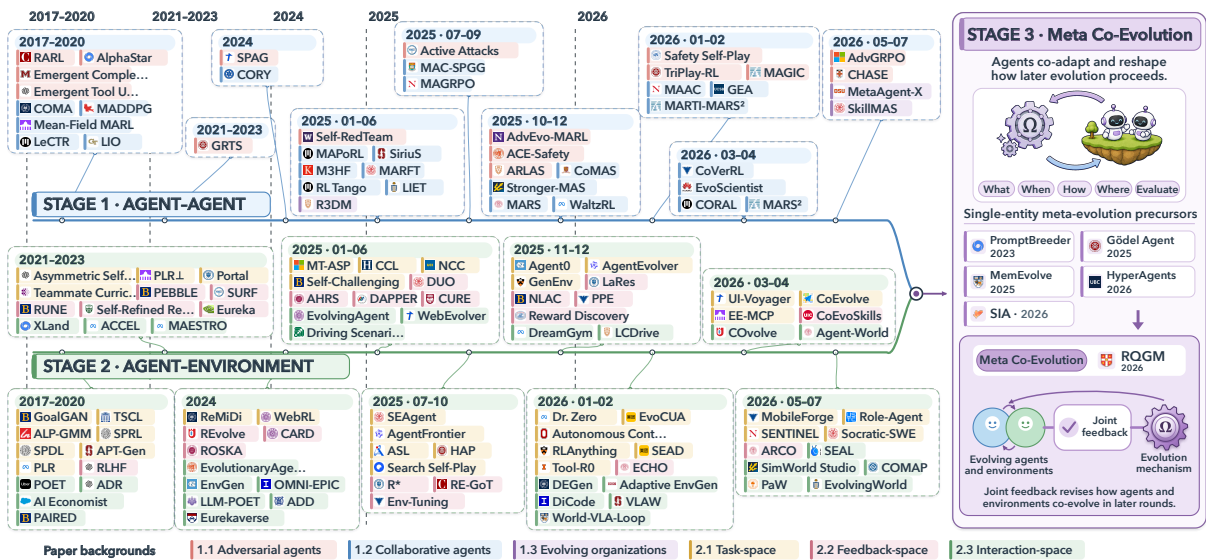


图 3：智能体系统中共同进化的论文分布图。

试图让另一个说出秘密词语，而另一个则试图猜出该词语。最近，这一思想被广泛应用于大语言模型（LLM）安全领域，攻击者智能体生成越狱提示，防御者智能体学习拒绝这些提示，双方相互促进改进。ACE-Safety（Li 等人，2025c）使用蒙特卡洛树搜索寻找攻击，AdvGRPO（Bullwinkel 等人，2026）通过密集多通道奖励稳定联合 GRPO，Yun 等人（2025）使已发现的攻击失效，从而迫使攻击者不断发现新的攻击类型，而不是坍塌到少数几种模式。MAGIC（Wen 等人，2026）运行多轮博弈，使攻击者能够发现单轮攻击无法发现的弱点，CHASE（Markasserithodi 等人，2026）放弃预设攻击模板，使其防御者对未见过的攻击保持鲁棒。一些工作还通过自我博弈将两种角色合并到一个模型中，其中 Self-RedTeam（Liu 等人，2025）使双方在线相互对抗，Wang 等人（2026a）重放过去的失败以避免遗忘困难案例。除了直接提示之外，ARLAS（Wang 等人，2025c）将博弈扩展到工具使用，将有害指令隐藏在智能体读取的工具输出中。

3.1.2 多源对抗压力

当智能体必须同时与多个对手保持竞争力时，会产生多源对抗压力。阿尔法星（AlphaStar）（Vinyals 等人，2019）通过实时战略游戏中的联赛训练提供了一个经典示例，主智能体与针对其弱点和历史版本的利用者共同进化

防止遗忘的版本。后续工作以不同方式扩展了这一思想。马等人（2024a）共同演化一个防御者来对抗多样化的红队模型群体，使其能够跨攻击风格泛化，而不是过拟合于单一攻击者。TriPlay-RL（谭等人，2026）共同演化三个智能体——攻击者、防御者和评估者，使攻击生成、安全响应和安全判断同步演化。AdvEvo-MARL（潘等人，2025）训练攻击者对抗整个任务智能体团队，这些智能体在执行任务的同时学习抵御攻击，因此安全性内建于群体之中，而非作为独立的防护措施。

3.2 协作智能体

与对抗性智能体不同，协作性智能体通过共享目标共同进化，其中一个智能体的改进会迫使其伙伴调整协调与构建的基础。本文区分两种结构：并行协作，其中对等智能体要么没有角色，要么可以承担任何角色；以及角色分化协作，其中固定且不同的角色为智能体提供了不同的改进路径。

3.2.1 并行协作

并行协作源于多智能体强化学习（Multi-Agent Reinforcement Learning, MARL），其中对等智能体共享单一任务奖励，并随着其他智能体的变化而协同适应（Lowe 等，2017；Foerster 等，2018；Yang 等，2018）。近期工作将这一思想引入大语言模型智能体。Liu 等（2026f）利用共享奖励和群体相对优势训练它们，而 Liu 等（2026e）则比较了联合历史与

局部历史批评家以稳定其共同进化。MARS² (Li et al., 2026a; Wang 等人, 2026b) 允许多个智能体在共享搜索树上共同进化, 将每个程序的测试奖励传播到整个树而非仅其最终节点。除最终结果外, 其他工作从协作过程本身汲取共同进化压力 (Omidshafiei 等人, 2019; Yang 等人, 2020)。一些方法将交互轨迹转化为奖励, 其中 CoMAS (Xue 等人, 2025) 和 MA-PoRL (Park 等人, 2025) 根据智能体的讨论给予奖励, 例如某轮发言是否纠正或说服了后续回应。另一些方法修订引导后续合作的信号, 其中 LIET (Li 等人, 2025b) 从消息中构建共享合作提示列表, M3HF (Wang 等人, 2025b) 将团队行为反馈转化为每个智能体的奖励, MAC-SPGG (Liang 等人, 2025) 使用序贯公共物品奖励以抑制搭便车行为。并行协作也可跨越独立工作空间, 智能体独立探索并分享所学, 如 CORAL (Qu 等人, 2026) 通过共享记忆扩散尝试、笔记和技能, GEA (Weng 等人, 2026) 将群体视为进化单元, 使一个智能体的执行日志、失败和改进指导其他智能体的更新。

3.2.2 角色差异化协作

角色差异化协作共同进化具有不同角色的智能体。常见模式是生成-修订循环。CORY (Ma 等人, 2024b) 是一种最小形式, 其中先驱者回答、观察者修订, 周期性的角色交换保持两者耦合。RL Tango (Zha 等人, 2025) 从结果正确性训练验证器以提供过程反馈, 而 CoVerRL (Pan 等人, 2026b) 在标签不可用时从多数投票答案引导其验证器。WaltzRL (Zhang 等人, 2025b) 联合训练对话智能体和批评智能体, 仅当其建议改善伙伴的下一轮回应时才给予奖励。其他方法按智能体能力拆分任务。EvoScientist (Lyu 等人, 2026) 让研究员和工程师智能体通过持久记忆跨任务积累经验。Sirius (Zhao 等人, 2025) 从成功和修复的协作轨迹中训练领域智能体, 如物理学家、数学家和总结者, 而 MARS (Chen 等人, 2026a) 通过共享轨迹奖励将系统 1 信息压缩与系统 2 推理耦合。最后, MARFT (Liao 等人, 2025) 和

Stronger-MAS (赵等人, 2026) 专注于信用分配, 将共享的团队奖励归因于每个角色, 以便每个角色都能改进。

3.3 演化中的智能体组织

超越固定角色, 智能体可以与组织共同演化, 包括角色分配和交互结构。R3DM (Goel 等人, 2025) 从智能体行为中发现角色, 并使其与智能体策略共同适应, 推动智能体表现出独特行为。SkillMAS (Pan 等人, 2026a) 联合更新智能体技能和团队结构, 当执行轨迹揭示不匹配时进行重构。MetaAgent-X (Zhang 等人, 2026c) 同时训练 workflow 设计器和执行器智能体, 使它们相互促进。

4 智能体-环境共同演化

本阶段研究智能体与自适应环境之间的共同演化 E 。随着智能体的改进, 环境重塑它们面临的任务、对其输出的反馈或它们行动的世界, 我们根据哪种变化主要驱动进一步的智能体演化来对方法进行分类。

4.1 任务空间共同演化

在任务空间共同演化中, 环境适应智能体接下来解决的问题。该文献有两种形式: 从现有任务池中选择任务和生成新的任务规范。

4.1.1 暴露与选择 暴露与选择方法保持任务空间固定, 并调整智能体所接触的任务。早期工作通过学习进度来调整暴露。对于离散任务或数据源, 课程学习 (Curriculum Learning) (Matiisen 等, 2020) 通过估计的进度调整采样权重, 而针对连续任务参数的方法 (Klink 等, 2019, 2020; Portelas 等, 2019; Fang 等, 2021) 将采样引导至智能体当前的能力范围, 这一思想后来被 Xu 等 (2025a) 扩展, 引入教师模型以提高学生仍然失败的任务的采样率。除了能力匹配之外, 基于回放的采样 (Jiang 等, 2021b, a) 重新访问仍具有学习潜力的过往关卡, 而基于遗憾的采样 (Beukman 等, 2024; Monette 等, 2025) 优先考虑智能体落后于更强参考的配置。在多智能体环境中, Bhati 等 (2023) 通过改变与不同能力队友的接触来控制任务难度。PORTAL (Wu 等, 2024)

选择能够引导智能体逐步达到困难目标的中间任务。对于大语言模型智能体，Wang 等（2026c）重新采样与智能体自身失败模式相匹配的任务，而 SEAD（Dai 等，2026）采样模拟用户画像，这些画像随着服务智能体的改进而变得更难处理。

4.1.2 自适应任务生成

生成式任务构建创造新任务，而非重新排列现有任务池，将其置于智能体当前能力前沿附近（Florensa 等，2018）。即使任务由智能体生成，它也属于环境演化，因为该智能体并非在同一任务内竞争或协作的同类。具身工作通过目标生成使这一点明确。OpenAI 等（2021）探索了具有挑战性但可达目标的环境，Kharyal 等（2025）通过多个教师扩大目标覆盖范围。CCL（Lin 等，2025）将其扩展到多智能体强化学习，并具有智能体特定的子任务。近期的大语言模型智能体方法在多个交互环境中实例化了这一循环。在工具使用中，生成器根据求解器的成功率校准任务难度（Guo 等，2025；Acikgoz 等，2026；Xia 等，2025；Chen 等，2025a），将其失败的轨迹转化为新任务（Wang 等，2026e；Yang 等，2026；Zhai 等，2025b），或将每个任务与验证函数配对（Zhou 等，2025）。在搜索任务中，提议者从检索行为生成多跳可验证问题（Lu 等，2025a；Sun 等，2025b；Yue 等，2026）。在图形用户界面和计算机使用中，一些方法探索不熟悉的软件以了解其支持的操作，并将其转化为任务（Liu 等，2026a；Xue 等，2026b；Sun 等，2025c），另一些方法从智能体的失败尝试中推导任务（Qi 等，2025），或通过扰动现有任务中的参数（如时间和数量）来生成任务（Lin 等，2026），还有一些方法（Xue 等，2026a；He 等，2026）生成每个任务运行所需的验证器和设置脚本。在软件工程中，Socratic-SWE（Xiao 等，2026）从其求解轨迹构建仓库修复任务。最后，RLAnything（Wang 等，2026d）在计算机使用、游戏和编码任务中联合重写任务、策略和奖励。

4.2 反馈空间协同演化

除了调整智能体面临的任务之外，环境还可以在如何评估、奖励或诊断智能体行为方面协同演化。

4.2.1 偏好驱动的反饋

反馈协同演化可能由不断演化的奖励模型驱动，该模型通过轨迹对上的偏好比较（Christiano et al., 2017）学习，并由脚本化教师而非人类进行评分。PEBBLE（Lee et al., 2021）在奖励模型变化时重新标记回放缓冲区中的轨迹，重用已存储的轨迹而非收集新标签。为进一步减少标注，Liang et al. (2022) 和 Zhu et al. (2025b) 寻找奖励模型最不确定的轨迹对，DAP-PER（Kadokawa et al., 2026）将来自不同策略的轨迹配对以扩大行为对比，DUO（Feng et al., 2025）则针对奖励分歧大且片段行为多样的轨迹对。

4.2.2 结果驱动的反饋

反馈协同演化也可能由任务结果驱动。在具身控制中（Song et al., 2023; Ma et al., 2024c; Sun et al., 2025a; Hazra et al., 2025），基于大语言模型的系统在候选奖励函数下训练策略，并根据轨迹回报或任务失败修订该函数。ROSKA（Huang et al., 2025a）和 LaRes（Li et al., 2026b）协同搜索奖励候选与策略变体，保留表现最佳的奖励，RE-GoT（Yao et al., 2026a）根据包括视觉评估在内的轨迹反馈细化分解后的奖励，Lu et al. (2025c) 则根据策略遗憾更新奖励参数。CURE（Wang et al., 2025a）和 CoEvoSkills（Zhang et al., 2026a）从执行失败中演化其单元测试或替代验证器。除了重写奖励或验证器，Lu et al. (2026b) 在训练早期添加纠正性提示，随后在智能体改进时撤回这些提示，而 AHRs（Huang et al., 2025b）根据固定奖励项当前对学习的贡献重新加权。

4.2.3 一致性增强的反饋

一致性约束在偏好或结果信号之上进一步增强了评估器学习。SURF（Park et al., 2022）通过重用奖励模型已自信排序的片段对来增强稀疏标签。R*（Li et al., 2025a）仅在多个评论者对同一片段排序达成一致时更新奖励参数。ARCO（Tian et al., 2026）与策略协同训练逐步评分的评分标准评估器，并要求这些步骤分数之和等于最终任务结果。NLAC（Hong et al., 2025）训练语言评论者，使其对后续步骤的预测与轨迹上的后续步骤保持一致，ECHO（Li et al., 2026c）更新

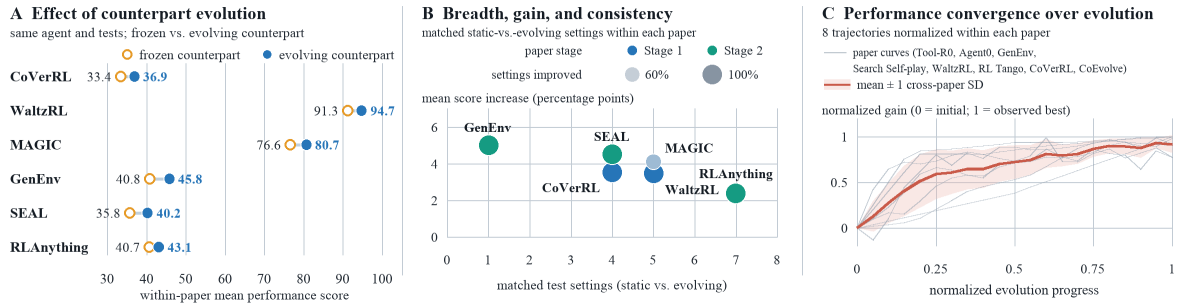


图 4：跨论文证据展示协同进化的效果、一致性与收敛性。数据选择与聚合细节见附录 C。

其诊断性批评者仅在采纳其建议确实能改进策略时才发挥作用。

4.3 交互空间协同进化

除任务与反馈外，环境亦可适应智能体行动所在的空间，或构建日益具有挑战性的可执行世界，或学习世界模型以替代真实环境。

4.3.1 可执行世界构建

在经典网格世界导航与连续控制中，POET (Wang 等, 2019) 演化环境-智能体种群，而基于遗憾的方法 (Dennis 等, 2020; Parker-Holder 等, 2022; Chung 等, 2024) 生成针对智能体弱点的关卡。近期，DE-Gen (Mead 等, 2025) 在学生探索过程中在线生成关卡部件，COvolve (Sygkounas 等, 2026) 利用大语言模型协同演化关卡与智能体控制器的代码。在机器人学与具身仿真中，ADR (OpenAI 等, 2019) 随策略改进扩大仿真随机化范围，后续工作则直接生成世界。OMNI-EPIC (Faldor 等, 2025) 依据趣味性筛选新环境，LLM-POET (Aki 等, 2024) 借助大语言模型生成器将 POET 适配至具身仿真。其他工作 (Liang 等, 2024; Abouelazm 等, 2025; Yeo 等, 2026) 生成难度随智能体学习潜力变化的环境，SimWorld Studio (Kang 等, 2026) 构建经物理与视觉校验的 3D 场景。在游戏与开放式世界中，XLand (Team 等, 2021) 根据智能体性能调整游戏规则与布局。MAESTRO (Samvelyan 等, 2023) 通过围绕不同合作玩家生成关卡，将基于遗憾的设计扩展至多智能体游戏。EnvGen (Zala 等, 2024) 重新生成针对智能体薄弱技能的环境，Di-

代码 (Code) (Mitsides 等, 2026) 构建关卡档案，从最高价值的关卡生成新关卡。对于工具使用智能体，智能体世界 (Agent-World) (Dong 等, 2026) 扩展工具组合和数据库状态以暴露新的能力缺口，而 SEAL (Hu 等, 2026) 从验证过的轨迹中细化环境的约束和恢复反馈，以纠正反复出现的智能体错误。最后，构建针对管理智能体群体的规则。Zheng 等 (2020) 学习一种税收政策，重塑经济智能体的激励，Li 等 (2024) 演化社会规范，跨代重塑智能体行为。

4.3.2 基于模型的世界构建

近期工作作用与智能体共同演化的世界模型替代真实环境。在基于文本的设置中，WebEvolver (Fang 等, 2025b) 将网络世界模型与智能体策略共同训练，作为虚拟网络服务器用于合成轨迹和前瞻。COMAP (Liu 等, 2026h) 使用在线文本世界模型预测未来反馈，指导动作细化。DreamGym (Chen 等, 2025b) 从经验模型合成转移，其回放缓冲区在训练期间增长。PaW (Lu 等, 2026a) 在智能体强化学习中添加下一观测预测作为辅助信号。在角色扮演仿真中，EvolvingWorld (Zong 等, 2026) 让世界模型决定跟踪哪些方面，并演化世界状态，随后塑造角色演化。在具身和视觉领域，世界模型减少昂贵的真实世界交互。EvolvingAgent (Feng 等, 2026) 从《我的世界》(Minecraft) 经验中改进世界模型和智能体规划器。Liu 等 (2026g) 和 Guo 等 (2026) 使用预测未来帧的视频世界模型训练智能体策略，两者共同改进。LCDrive (Tan 等, 2025) 通过闭环强化学习优化潜在世界模型令牌与驾驶动作。

通过闭环强化学习。

5 元共同演化

5.1 走向开放式

如图 4 所示，阶段 1 和阶段 2 在大多数设置下均提升了性能，但随着演化接近平台期，增益逐渐减小。通过允许演化机制本身发生变化，元协同演化可能突破这一瓶颈，开辟新的改进方向。元协同演化提供了一条通往开放式演化 (Stanley et al., 2017; Hughes et al., 2024) 的路径，这要求持续的新颖性 ($\Omega^{t+1} \neq \Omega^t$)，因此机制更新能够不断开启新的可学习方向，以及无界发散 ($\lim_{t \rightarrow \infty} H(S^t, \Omega^t) = \infty$)，使得系统的适应能力没有有限的上界。从这个意义上说，元协同演化不仅仅是另一个类别，而是从设计规则下的适应迈向自我扩展的智能体系统的一步。我们将演化机制分解为五个自适应决策：(1) 演化什么，选择适应目标，例如智能体骨干、组织架构或环境工件，如任务、奖励和世界；(2) 何时演化，在失败、平台期或分布偏移后触发更新；(3) 如何演化，通过骨干训练、组织架构修订或结构生成与编辑产生变体；(4) 何处演化，涵盖演化领域，如工具使用、网络或机器人技术，以及其运行环境，如沙盒、模拟或真实环境；(5) 如何评估，通过性能、新颖性、安全性或鲁棒性来评判演化质量，从而塑造后续演化的方向。这一标准将元协同演化与孤立的元演化区分开来。机制修订必须由低层协同演化生态系统的联合轨迹所引发，并且必须随后改变该生态系统后续的演化条件。

5.2 从前体到元协同演化

大多数相关方法仍停留在单一实体的前驱阶段。提示词培育器 (PromptBreeder) (Fernando 等人, 2024) 演化变异提示，以控制未来任务提示的变化方式。哥德尔智能体 (Gödel Agent) (Yin 等人, 2025) 与超智能体 (HyperAgents) (Zhang 等人, 2026b) 演化智能体自身的自我修改机制。记忆演化 (MemEvolve) (Zhang 等人, 2025a) 演化管理经验重用的记忆架构，而自我改进智能体 (SIA) (Hebbbar 等人, 2026) 则利用轨迹

轨迹反馈用于在参数更新与权重更新之间进行选择。这些方法使进化机制本身具备可进化性，但缺乏底层协同进化系统。RQGM (Iacob 等人, 2026) 更进一步，通过协同进化任务智能体与评估器，而其元智能体利用联合反馈指导后续进化。

6 挑战与未来方向

动态评估。现有基准主要衡量智能体在工具使用 (Yao et al., 2024; Lu et al., 2025b; Liu et al., 2026b,c,d)、网页浏览 (Zhou et al., 2024a; Wei et al., 2025)、软件工程 (Jimenez et al., 2024; Miserendino et al., 2025)、计算机使用 (Trivedi et al., 2024; Xie et al., 2024; Merrill et al., 2026) 以及多智能体交互 (Zhou et al., 2024b; Mou et al., 2025; Zhu et al., 2025a) 等领域的最终能力。PostTrain-Bench (Rank et al., 2026) 通过测试智能体能否自主后训练基础大语言模型，向进化感知评估迈出了一步。然而，评估共同进化需要更进一步：必须确定所有进化组件是否都得到改进，它们的收益是否能迁移到未见过的伙伴和环境，以及每个组件对联合进展的贡献。由于更高的任务成功率可能掩盖剥削行为 (Thaman, 2026)，此类系统可能因评估器剥削、伙伴过拟合或多样性崩溃而失败。因此，评估应将固定基准与过程级测试相结合，例如历史交叉对战、组件消融和留出评估器。

扩展共同进化。当前系统主要研究局部循环，例如攻击者-防御者训练 (Wen et al., 2026)、策略-奖励适应 (Lu et al., 2025c) 或智能体-任务生成 (Acikgoz et al., 2026)。未来工作可以将这些扩展到智能体、其框架和环境共同变化的系统。挑战不在于使更多组件具有适应性，而在于决定哪些组件应该改变，它们的更新如何相互影响，以及如何保持共同进化压力富有成效，而不是不稳定或被单一组件主导。元共同进化通过让系统自行做出这些进化决策，开始应对这一挑战，超越了人类预定义的改进路径。

安全与治理。随着共同进化变得更加自主，人类可能失去对系统行为的控制。进化中的智能体可能发展出攻击策略、工具使用模式、通信协议（Tucker 等，2022；Motwani 等，2024）或超出人类理解且难以监控的组织行为。在元共同进化中风险更为突出，系统还可能改变哪些行为得到奖励和保留。因此，未来工作需要建立治理机制，使开放式进化保持可审计和可中断，包括沙箱部署、持续监控、回滚到已验证状态以及人工干预点。

7 结论

本综述聚焦于智能体系统中的共同进化，其中多个组件相互施加进化压力。我们通过一个渐进式分类法来组织该领域，其自适应边界从智能体本身扩展到智能体与其环境，最后到支配它们联合进化的机制，反映了人工干预的逐步减少。这一视角表明，未来的进展不在于更强的静态智能体，而在于通过共同进化持续改进的智能体，并辅以评估和治理来保持过程可靠可控。

局限性

元共同进化仍处于早期阶段。目前只有有限的工作符合我们对第三阶段的定义。因此，我们的讨论大多借鉴单实体元进化作为前身，这表明进化机制本身可以进化，但并未将这种变化与低层共同进化系统耦合。

安全与治理尚未操作化。我们将安全、监控和人工监督视为日益自主的共同进化的首要关注点，并指出共同进化系统特有的失效模式，如评估器利用、伙伴过拟合和多样性崩溃。但我们仅在需求层面处理这些问题，并未为其制定具体的保障措施或协议。

伦理声明

本文对智能体系统中的协同进化进行了全面综述，特别关注

关于智能体与其环境如何相互适应性地重塑彼此，从人类设计走向自我导向的演化。本综述中评述的所有研究工作均已恰当引用。据我们所知，所引用的材料均可公开获取，或依据允许用于研究综述的许可证提供。我们未进行额外的数据集整理或人工标注工作。因此，我们认为本文不引发任何伦理问题。

参考文献

- Ahmed Abouelazm, Tim Weinstein, Tim Joseph, Philip Schörner, and J. Marius Zöllner. 2025. [Automatic curriculum learning for driving scenarios: Towards robust and efficient reinforcement learning](#). In *IEEE Intelligent Vehicles Symposium, IV 2025, Cluj-Napoca, Romania, June 22-25, 2025*, pages 2333–2340. IEEE.
- Emre Can Acikgoz, Cheng Qian, Jonas Hübötter, Heng Ji, Dilek Hakkani-Tür, and Gokhan Tur. 2026. [Tool-r0: Self-evolving LLM agents for tool-learning from zero data](#). *CoRR*, abs/2602.21320.
- Fuma Aki, Riku Ikeda, Takumi Saito, Ciaran Regan, and Mizuki Oka. 2024. [LLM-POET: evolving complex environments using large language models](#). In *Proceedings of the Genetic and Evolutionary Computation Conference Companion, GECCO 2024, Melbourne, VIC, Australia, July 14-18, 2024*, pages 243–246. ACM.
- Bowen Baker, Ingmar Kanitscheider, Todor M. Markov, Yi Wu, Glenn Powell, Bob McGrew, and Igor Mordatch. 2020. [Emergent tool use from multi-agent autotutorials](#). In *8th International Conference on Learning Representations, ICLR 2020, Addis Ababa, Ethiopia, April 26-30, 2020*. OpenReview.net.
- Trapit Bansal, Jakub Pachocki, Szymon Sidor, Ilya Sutskever, and Igor Mordatch. 2018. [Emergent complexity via multi-agent competition](#). In *6th International Conference on Learning Representations, ICLR 2018, Vancouver, BC, Canada, April 30 - May 3, 2018, Conference Track Proceedings*. OpenReview.net.
- Michael Beukman, Samuel Coward, Michael T. Matthews, Mattie Fellows, Minqi Jiang, Michael D. Dennis, and Jakob Nicolaus Foerster. 2024. [Refining minimax regret for unsupervised environment design](#). In *Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21-27, 2024*, volume 235 of *Proceedings of Machine Learning Research*, pages 3637–3657. PMLR / OpenReview.net.
- Rupali Bhati, Sai Krishna Gottipati, Clodéric Mars, and Matthew E. Taylor. 2023. [Curriculum learning for cooperation in multi-agent reinforcement learning](#). *CoRR*, abs/2312.11768.

- Blake Bullwinkel, Eugenia Kim, Amanda J. Minnich, and Mark Russinovich. 2026. [Learning to attack and defend: Adaptive red teaming of language models via GRPO](#). *CoRR*, abs/2606.09701.
- Dikshit Chauhan, Bapi Dutta, Indu Bala, Niki van Stein, Thomas Bäck, and Anupam Yadav. 2025. [Evolutionary computation and large language models: A survey of methods, synergies, and applications](#). *CoRR*, abs/2505.15741.
- Guoxin Chen, Zile Qiao, Wenqing Wang, Donglei Yu, Xuanzhong Chen, Hao Sun, Minpeng Liao, Kai Fan, Yong Jiang, Penguin Xie, Wayne Xin Zhao, Ruihua Song, and Fei Huang. 2026a. [Mars: Co-evolving dual-system deep research via multi-agent reinforcement learning](#). *Preprint*, arXiv:2510.04935.
- Mingguang Chen, Licheng Wang, and Bo Qu. 2026b. [Recursive self-improvement in AI: from bounded self-refinement to autonomous research loops](#). *CoRR*, abs/2607.07663.
- Xuanzhong Chen, Zile Qiao, Guoxin Chen, Liangcai Su, Zhen Zhang, Xinyu Wang, Pengjun Xie, Fei Huang, Jingren Zhou, and Yong Jiang. 2025a. [Agentfrontier: Expanding the capability frontier of LLM agents with zpd-guided data synthesis](#). *CoRR*, abs/2510.24695.
- Zhaorun Chen, Zhuokai Zhao, Kai Zhang, Bo Liu, Qi Qi, Yifan Wu, Tarun Kalluri, Sara Cao, Yuanhao Xiong, Haibo Tong, Huaxiu Yao, Hengduo Li, Jiacheng Zhu, Xian Li, Dawn Song, Bo Li, Jason Weston, and Dat Huynh. 2025b. [Scaling agent learning via experience synthesis](#). *CoRR*, abs/2511.03773.
- Pengyu Cheng, Tianhao Hu, Han Xu, Zhisong Zhang, Yong Dai, Lei Han, Nan Du, and Xiaolong Li. 2024. [Self-playing adversarial language game enhances LLM reasoning](#). In *Advances in Neural Information Processing Systems 37: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*.
- Paul F. Christiano, Jan Leike, Tom B. Brown, Miljan Martic, Shane Legg, and Dario Amodei. 2017. [Deep reinforcement learning from human preferences](#). In *Advances in Neural Information Processing Systems 30: Annual Conference on Neural Information Processing Systems 2017, December 4-9, 2017, Long Beach, CA, USA*, pages 4299–4307.
- Hojun Chung, Junseo Lee, Minsoo Kim, Dohyeong Kim, and Songhwai Oh. 2024. [Adversarial environment design via regret-guided diffusion models](#). In *Advances in Neural Information Processing Systems 37: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*.
- Yueqin Dai, Ning Gao, Wei Zhang, Jie Wang, Zichen Luo, Jinpeng Wang, Ruiyuan Wu, and Chaozheng Wang. 2026. [SEAD: self-evolving agent for multi-turn service dialogue](#). In *Findings of the Association for Computational Linguistics, ACL 2026, San Diego, California, United States, July 2-7, 2026*, pages 3674–3684. Association for Computational Linguistics.
- Michael Dennis, Natasha Jaques, Eugene Vinitzky, Alexandre M. Bayen, Stuart Russell, Andrew Critch, and Sergey Levine. 2020. [Emergent complexity and zero-shot transfer via unsupervised environment design](#). In *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual*.
- Guanting Dong, Juntong Lu, Junjie Huang, Wanjun Zhong, Longxiang Liu, Shijue Huang, Zhenyu Li, Yang Zhao, Xiaoshuai Song, Xiaoxi Li, Jiajie Jin, Yutao Zhu, Hanbin Wang, Fangyu Lei, Qinyu Luo, Mingyang Chen, Zehui Chen, Jiazhan Feng, Ji-Rong Wen, and Zhicheng Dou. 2026. [Agent-world: Scaling real-world environment synthesis for evolving general agent intelligence](#). *CoRR*, abs/2604.18292.
- Pengfei Du. 2026. [Memory for autonomous LLM agents: mechanisms, evaluation, and emerging frontiers](#). *CoRR*, abs/2603.07670.
- Maxence Faldor, Jenny Zhang, Antoine Cully, and Jeff Clune. 2025. [OMNI-EPIC: open-endedness via models of human notions of interestingness with environments programmed in code](#). In *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net.
- Jinyuan Fang, Yanwen Peng, Xi Zhang, Yingxu Wang, Xinhao Yi, Guibin Zhang, Yi Xu, Bin Wu, Siwei Liu, Zihao Li, Zhaochun Ren, Nikos Aletras, Xi Wang, Han Zhou, and Zaiqiao Meng. 2025a. [A comprehensive survey of self-evolving AI agents: A new paradigm bridging foundation models and lifelong agentic systems](#). *CoRR*, abs/2508.07407.
- Kuan Fang, Yuke Zhu, Silvio Savarese, and Li Fei-Fei. 2021. [Adaptive procedural task generation for hard-exploration problems](#). In *9th International Conference on Learning Representations, ICLR 2021, Virtual Event, Austria, May 3-7, 2021*. OpenReview.net.
- Tianqing Fang, Hongming Zhang, Zhisong Zhang, Kaixin Ma, Wenhao Yu, Haitao Mi, and Dong Yu. 2025b. [Webevolver: Enhancing web agent self-improvement with co-evolving world model](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, EMNLP 2025, Suzhou, China, November 4-9, 2025*, pages 8959–8975. Association for Computational Linguistics.
- Tongtong Feng, Xin Wang, Zekai Zhou, Ren Wang, Yuwei Zhan, Guangyao Li, Qing Li, and Wenwu Zhu. 2026. [Evolvingagent: Curriculum self-evolving agent with continual world model for long-horizon tasks](#). *Preprint*, arXiv:2502.05907.
- Xuening Feng, Zhaohui Jiang, Timo Kaufmann, Puchen Xu, Eyke Hüllermeier, Paul Weng, and Yifei Zhu. 2025. [DUO: diverse, uncertain, on-policy query generation and selection for reinforcement learning from](#)

- human feedback. In *Thirty-Ninth AAAI Conference on Artificial Intelligence, Thirty-Seventh Conference on Innovative Applications of Artificial Intelligence, Fifteenth Symposium on Educational Advances in Artificial Intelligence, AAAI 2025, Philadelphia, PA, USA, February 25 - March 4, 2025*, pages 16604–16612. AAAI Press.
- Chrisantha Fernando, Dylan Banarse, Henryk Michalewski, Simon Osindero, and Tim Rocktäschel. 2024. [Promptbreeder: Self-referential self-improvement via prompt evolution](#). In *Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21-27, 2024*, volume 235 of *Proceedings of Machine Learning Research*, pages 13481–13544. PMLR / OpenReview.net.
- Carlos Florensa, David Held, Xinyang Geng, and Pieter Abbeel. 2018. [Automatic goal generation for reinforcement learning agents](#). In *Proceedings of the 35th International Conference on Machine Learning, ICML 2018, Stockholmsmässan, Stockholm, Sweden, July 10-15, 2018*, volume 80 of *Proceedings of Machine Learning Research*, pages 1514–1523. PMLR.
- Jakob N. Foerster, Gregory Farquhar, Triantafyllos Afouras, Nantas Nardelli, and Shimon Whiteson. 2018. [Counterfactual multi-agent policy gradients](#). In *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence, (AAAI-18), the 30th Innovative Applications of Artificial Intelligence (IAAI-18), and the 8th AAAI Symposium on Educational Advances in Artificial Intelligence (EAAI-18), New Orleans, Louisiana, USA, February 2-7, 2018*, pages 2974–2982. AAAI Press.
- Huan-ang Gao, Jiayi Geng, Wenyue Hua, Mengkang Hu, Xinzhe Juan, Hongzhang Liu, Shilong Liu, Jiahao Qiu, Xuan Qi, Qihan Ren, Yiran Wu, Hongru Wang, Han Xiao, Yuhang Zhou, Shaokun Zhang, Jiayi Zhang, Jinyu Xiang, Yixiong Fang, Qiwen Zhao, and 8 others. 2026. [A survey of self-evolving agents: What, when, how, and where to evolve on the path to artificial super intelligence](#). *Trans. Mach. Learn. Res.*, 2026.
- Harsh Goel, Mohammad Omama, Behdad Chalaki, Vaishnav Tadiparthi, Ehsan Moradi-Pari, and Sandeep P. Chinchali. 2025. [R3DM: enabling role discovery and diversity through dynamics models in multi-agent reinforcement learning](#). In *Forty-second International Conference on Machine Learning, ICML 2025, Vancouver, BC, Canada, July 13-19, 2025*, volume 267 of *Proceedings of Machine Learning Research*. PMLR / OpenReview.net.
- Ian J. Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron C. Courville, and Yoshua Bengio. 2014. [Generative adversarial networks](#). *CoRR*, abs/1406.2661.
- Jiacheng Guo, Ling Yang, Peter Chen, Qixin Xiao, Yinjie Wang, Xinzhe Juan, Jiahao Qiu, Ke Shen, and Mengdi Wang. 2025. [Genenv: Difficulty-aligned co-evolution between LLM agents and environment simulators](#). *CoRR*, abs/2512.19682.
- Taicheng Guo, Xiuying Chen, Yaqi Wang, Ruidi Chang, Shichao Pei, Nitesh V. Chawla, Olaf Wiest, and Xi-angliang Zhang. 2024. [Large language model based multi-agents: A survey of progress and challenges](#). In *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence, IJCAI 2024, Jeju, South Korea, August 3-9, 2024*, pages 8048–8057. ijcai.org.
- Yanjiang Guo, Tony Lee, Lucy Xiaoyang Shi, Jianyu Chen, Percy Liang, and Chelsea Finn. 2026. [VLAW: iterative co-improvement of vision-language-action policy and world model](#). *CoRR*, abs/2602.12063.
- Rishi Hazra, Alkis Sygkounas, Andreas Persson, Amy Loutfi, and Pedro Zuidberg Dos Martires. 2025. [Re-volve: Reward evolution with large language models using human feedback](#). In *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net.
- Tiantian He, Yihang Chen, Keyue Jiang, Ka Yiu Lee, Kaiwen Zhou, Kun Shao, and Shuai Wang. 2026. [EE-MCP: self-evolving MCP-GUI agents via automated environment generation and experience learning](#). *CoRR*, abs/2604.09815.
- Prannay Hebbar, Yogendra Manawat, Samuel Verboomen, Alesia Ivanova, Selvam Palanimalai, Kunal Bhatia, and Vignesh Baskaran. 2026. [SIA: self improving AI with harness & weight updates](#). *CoRR*, abs/2605.27276.
- W Daniel Hillis. 1990. [Co-evolving parasites improve simulated evolution as an optimization procedure](#). *Physica D: Nonlinear Phenomena*, 42(1-3):228–234.
- Joey Hong, Kang Liu, Zhan Ling, Jiecao Chen, and Sergey Levine. 2025. [Natural language actor-critic: Scalable off-policy learning in language space](#). *CoRR*, abs/2512.04601.
- Yihao Hu, Zhihao Wen, Xiu Jin Liu, Pan Wang, Xin Zhang, and Wei Wu. 2026. [SEAL: synergistic co-evolution of agents and learning environments](#). *CoRR*, abs/2605.24426.
- Changxin Huang, Yanbin Chang, Junfan Lin, Junyang Liang, Runhao Zeng, and Jianqiang Li. 2025a. [Efficient language-instructed skill acquisition via reward-policy co-evolution](#). In *Thirty-Ninth AAAI Conference on Artificial Intelligence, Thirty-Seventh Conference on Innovative Applications of Artificial Intelligence, Fifteenth Symposium on Educational Advances in Artificial Intelligence, AAAI 2025, Philadelphia, PA, USA, February 25 - March 4, 2025*, pages 14576–14584. AAAI Press.
- Changxin Huang, Junyang Liang, Yanbin Chang, Jingzhao Xu, and Jianqiang Li. 2025b. [Automated hybrid reward scheduling via large language models for robotic skill learning](#). In *IEEE International*

- Conference on Robotics and Automation, ICRA 2025, Atlanta, GA, USA, May 19-23, 2025*, pages 8242–8248. IEEE.
- Yuchen Huang, Sijia Li, Minghao Liu, Wei Liu, Shijue Huang, Zhiyuan Fan, Hou Pong Chan, and Yi R. Fung. 2025c. [Environment scaling for interactive agentic experience collection: A survey](#). *CoRR*, abs/2511.09586.
- Edward Hughes, Michael D. Dennis, Jack Parker-Holder, Feryal M. P. Behbahani, Aditi Mavalankar, Yuge Shi, Tom Schaul, and Tim Rocktäschel. 2024. [Position: Open-endedness is essential for artificial superhuman intelligence](#). In *Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21-27, 2024*, volume 235 of *Proceedings of Machine Learning Research*, pages 20597–20616. PMLR / OpenReview.net.
- Alex Jacob, Andrej Jovanovic, William F. Shen, Daniel Burkhardt, Meghdad Kurmanji, Nurbek Tastan, Lorenzo Sani, Niccolò Alberto Elia Venanzi, Ambroise Odonnat, Zeyu Cao, Bill Marino, Xinchu Qiu, and Nicholas D. Lane. 2026. [The red queen gödel machine: Co-evolving agents and their evaluators](#). *CoRR*, abs/2606.26294.
- Minqi Jiang, Michael Dennis, Jack Parker-Holder, Jakob N. Foerster, Edward Grefenstette, and Tim Rocktäschel. 2021a. [Replay-guided adversarial environment design](#). In *Advances in Neural Information Processing Systems 34: Annual Conference on Neural Information Processing Systems 2021, NeurIPS 2021, December 6-14, 2021, virtual*, pages 1884–1897.
- Minqi Jiang, Edward Grefenstette, and Tim Rocktäschel. 2021b. [Prioritized level replay](#). In *Proceedings of the 38th International Conference on Machine Learning, ICML 2021, 18-24 July 2021, Virtual Event*, volume 139 of *Proceedings of Machine Learning Research*, pages 4940–4950. PMLR.
- Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik R. Narasimhan. 2024. [Swe-bench: Can language models resolve real-world github issues?](#) In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net.
- Yuki Kadokawa, Jonas Frey, Takahiro Miki, Takamitsu Matsubara, and Marco Hutter. 2026. [DAPPER: discriminability-aware policy-to-policy preference-based reinforcement learning for query-efficient robot skill acquisition](#). *IEEE Robotics Autom. Mag.*, 33(1):151–166.
- Haoqiang Kang, Xiaokang Ye, Yuhang Liu, Sidhant Hitesh Mantri, Lingjun Mao, James Fleming, Drishti Regmi, and Lianhui Qin. 2026. [Simworld studio: Automatic environment generation with evolving coding agent for embodied agent learning](#). *CoRR*, abs/2605.09423.
- Chaitanya Kharyal, Sai Krishna Gottipati, Tanmay Kumar Sinha, Fatemeh Abdollahi, Srijita Das, and Matthew E. Taylor. 2025. [Do as you teach: a multi-teacher approach to self-play in deep reinforcement learning](#). *Neural Comput. Appl.*, 37(8):5945–5956.
- Pascal Klink, Hany Abdulsamad, Boris Belousov, and Jan Peters. 2019. [Self-paced contextual reinforcement learning](#). In *3rd Annual Conference on Robot Learning, CoRL 2019, Osaka, Japan, October 30 - November 1, 2019, Proceedings*, volume 100 of *Proceedings of Machine Learning Research*, pages 513–529. PMLR.
- Pascal Klink, Carlo D’Eramo, Jan Peters, and Joni Pajarinen. 2020. [Self-paced deep reinforcement learning](#). In *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual*.
- Kimin Lee, Laura M. Smith, and Pieter Abbeel. 2021. [PEBBLE: feedback-efficient interactive reinforcement learning via relabeling experience and unsupervised pre-training](#). In *Proceedings of the 38th International Conference on Machine Learning, ICML 2021, 18-24 July 2021, Virtual Event*, volume 139 of *Proceedings of Machine Learning Research*, pages 6152–6163. PMLR.
- Pengfei Li, Shijie Wang, Fangyuan Li, Yikun Fu, Kaifeng Liu, Kaiyan Zhang, Dazhi Zhang, Yuqiang Li, Biqing Qi, and Bowen Zhou. 2026a. [Mars²: Scaling multi-agent tree search via reinforcement learning for code generation](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2026, San Diego, California, United States, July 2-7, 2026*, pages 33319–33335. Association for Computational Linguistics.
- Pengyi Li, Jianye Hao, Hongyao Tang, Yifu Yuan, Jinbin Qiao, Zibin Dong, and Yan Zheng. 2025a. [R*: Efficient reward design via reward structure evolution and parameter alignment optimization with large language models](#). In *Forty-second International Conference on Machine Learning, ICML 2025, Vancouver, BC, Canada, July 13-19, 2025*, volume 267 of *Proceedings of Machine Learning Research*. PMLR / OpenReview.net.
- Pengyi Li, Hongyao Tang, Jinbin Qiao, Yan Zheng, and Jianye Hao. 2026b. [Lares: evolutionary reinforcement learning with llm-based adaptive reward search](#). *Advances in Neural Information Processing Systems*, 38:23270–23298.
- Shimin Li, Tianxiang Sun, and Xipeng Qiu. 2024. [Agent alignment in evolving social norms](#). *CoRR*, abs/2401.04620.
- Xinran Li, Chenjia Bai, Zijian Li, Jiakun Zheng, Ting Xiao, and Jun Zhang. 2025b. [Learn as individuals, evolve as a team: Multi-agent llms adaptation in embodied environments](#). *CoRR*, abs/2506.07232.

- Xurui Li, Kaisong Song, Rui Zhu, Pin-Yu Chen, and Haixu Tang. 2025c. [Adversarial attack-defense co-evolution for LLM safety alignment via tree-group dual-aware search and optimization](#). *CoRR*, abs/2511.19218.
- Zhicong Li, Lingjie Jiang, Yulan Hu, Xingchen Zeng, Yixia Li, Xiangwen Zhang, Guanhua Chen, Zheng Pan, Xin Li, and Yong Liu. 2026c. [No more stale feedback: Co-evolving critics for open-world agent learning](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12643–12660, San Diego, California, United States. Association for Computational Linguistics.
- William Liang, Sam Wang, Hung-Ju Wang, Osbert Bastani, Dinesh Jayaraman, and Yecheng Jason Ma. 2024. [Eurekaverse: Environment curriculum generation via large language models](#). *CoRR*, abs/2411.01775.
- Xinran Liang, Katherine Shu, Kimin Lee, and Pieter Abbeel. 2022. [Reward uncertainty for exploration in preference-based reinforcement learning](#). In *The Tenth International Conference on Learning Representations, ICLR 2022, Virtual Event, April 25-29, 2022*. OpenReview.net.
- Yunhao Liang, Yuan Qu, Jingyuan Yang, Shaochong Lin, and Zuo-Jun Max Shen. 2025. [Everyone contributes! incentivizing strategic cooperation in multi-llm systems via sequential public goods games](#). *CoRR*, abs/2508.02076.
- Junwei Liao, Muning Wen, Jun Wang, and Weinan Zhang. 2025. [MARFT: multi-agent reinforcement fine-tuning](#). *CoRR*, abs/2504.16129.
- Yufei Lin, Chengwei Ye, Huanzhen Zhang, Kangsheng Wang, Linuo Xu, Shuyan Liu, and Zeyu Zhang. 2025. [CCL: collaborative curriculum learning for sparse-reward multi-agent reinforcement learning via co-evolutionary task evolution](#). In *Advanced Intelligent Computing Technology and Applications - 21st International Conference, ICIC 2025, Ningbo, China, July 26-29, 2025, Proceedings, Part XII*, volume 15853 of *Lecture Notes in Computer Science*, pages 51–62. Springer.
- Zichuan Lin, Feiyu Liu, Yijun Yang, Jiafei Lyu, Yiming Gao, Yicheng Liu, Zhicong Lu, Yangbin Yu, Mingyu Yang, Junyou Li, Deheng Ye, and Jie Jiang. 2026. [Ui-voyager: A self-evolving GUI agent learning via failed experience](#). *CoRR*, abs/2603.24533.
- Guangyi Liu, Pengxiang Zhao, Gao Wu, Yiwen Yin, Mading Li, Liang Liu, Congxiao Liu, Zhang Qi, Mengyan Wang, Liang Guo, and Yong Liu. 2026a. [Mobileforge: Annotation-free adaptation for mobile GUI agents with hierarchical feedback-guided policy optimization](#). *CoRR*, abs/2606.19930.
- Jiayu Liu, Qihan Lin, Cheng Qian, Rui Wang, Emre Can Acikgoz, Xiaocheng Yang, Jiateng Liu, Zhenhailong Wang, Xiushi Chen, Heng Ji, and Dilek Hakkani-Tür. 2026b. [Planbench-xl: Evaluating long-horizon planning of llm tool-use agents in large-scale tool ecosystems](#). *arXiv preprint arXiv:2606.22388*.
- Jiayu Liu, Cheng Qian, Zhaochen Su, Qing Zong, Shijue Huang, Bingxiang He, and Yi R. Fung. 2026c. [CostBench: Evaluating multi-turn cost-optimal planning and adaptation in dynamic environments for LLM tool-use agents](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12826–12858, San Diego, California, United States. Association for Computational Linguistics.
- Jiayu Liu, Cheng Qian, Zhenhailong Wang, Bingxuan Li, Jiateng Liu, Qing Zong, Heng Wang, Jeonghwan Kim, Yumeng Wang, Bingxiang He, and 1 others. 2026d. [Adaplanbench: Evaluating adaptive planning in large language model agents under world and user constraints](#). *arXiv preprint arXiv:2606.05622*.
- Mickel Liu, Liwei Jiang, Yancheng Liang, Simon Shaolei Du, Yejin Choi, Tim Althoff, and Natasha Jaques. 2025. [Chasing moving targets with online self-play reinforcement learning for safer language models](#). *CoRR*, abs/2506.07468.
- Shuo Liu, Tianle Chen, Ryan Amiri, and Christopher Amato. 2026e. [Learning decentralized LLM collaboration with multi-agent actor critic](#). *CoRR*, abs/2601.21972.
- Shuo Liu, Zeyu Liang, Xueguang Lyu, and Christopher Amato. 2026f. [LLM collaboration with multi-agent reinforcement learning](#). In *Fortieth AAAI Conference on Artificial Intelligence, Thirty-Eighth Conference on Innovative Applications of Artificial Intelligence, Sixteenth Symposium on Educational Advances in Artificial Intelligence, AAAI 2026, Singapore, January 20-27, 2026*, pages 32150–32158. AAAI Press.
- Xiaokang Liu, Zechen Bai, Hai Ci, Kevin Yuchen Ma, and Mike Zheng Shou. 2026g. [World-vla-loop: Closed-loop learning of video world model and VLA policy](#). *CoRR*, abs/2602.06508.
- Youwei Liu, Jian Wang, Hanlin Wang, and Wenjie Li. 2026h. [COMAP: co-evolving world models and agent policies for LLM agents](#). *CoRR*, abs/2606.02372.
- Yuxuan Liu, Zhaochen Su, Lingyun Xie, Yuhao Zhang, Qing Zong, Jiahe Guo, Zhongwei Xie, Yiyang Ji, Yauwai Yim, Hongyu Luo, Xiyu Ren, Ruan Chenyu, Haoran Li, and Yangqiu Song. 2026i. [Skillrevise: Improving llm-authored agent skills via trace-conditioned skill revision](#). *CoRR*, abs/2606.01139.
- Ryan Lowe, Yi Wu, Aviv Tamar, Jean Harb, Pieter Abbeel, and Igor Mordatch. 2017. [Multi-agent actor-critic for mixed cooperative-competitive environments](#). In *Advances in Neural Information Processing Systems 30: Annual Conference on Neural Information Processing Systems 2017, December 4-9, 2017, Long Beach, CA, USA*, pages 6379–6390.

- Hongliang Lu, Yuhang Wen, Pengyu Cheng, Ruijin Ding, Haotian Xu, Jiaqi Guo, Chutian Wang, Haonan Chen, Xiaoxi Jiang, and Guanjuan Jiang. 2025a. [Search self-play: Pushing the frontier of agent capability without supervision](#). *CoRR*, abs/2510.18821.
- Jiarui Lu, Thomas Holleis, Yizhe Zhang, Bernhard Aumayer, Feng Nan, Haoping Bai, Shuang Ma, Shen Ma, Mengyu Li, Guoli Yin, Zirui Wang, and Ruoming Pang. 2025b. [Toolsandbox: A stateful, conversational, interactive evaluation benchmark for LLM tool use capabilities](#). In *Findings of the Association for Computational Linguistics: NAACL 2025, Albuquerque, New Mexico, USA, April 29 - May 4, 2025*, volume NAACL 2025 of *Findings of ACL*, pages 1160–1183. Association for Computational Linguistics.
- Ning Lu, Baijiong Lin, Shengcai Liu, Jiahao Wu, Haoze Lv, Yanbin Wei, Lingting Zhu, Shengju Qian, Xin Wang, Ying-Cong Chen, Qi Wang, and Ke Tang. 2026a. [Policy and world modeling co-training for language agents](#). *CoRR*, abs/2606.02388.
- Renzhi Lu, Zonghe Shao, Yuemin Ding, Ruijuan Chen, Dongrui Wu, Housheng Su, Tao Yang, Fumin Zhang, Jun Wang, Yang Shi, and 1 others. 2025c. [Discovery of the reward function for embodied reinforcement learning agents](#). *Nature Communications*, 16(1):11064.
- Siyuan Lu, Zechuan Wang, Hongxuan Zhang, Qintong Wu, Leilei Gan, Chenyi Zhuang, Jinjie Gu, and Tao Lin. 2026b. [Don't just fine-tune the agent, tune the environment](#). *Preprint*, arXiv:2510.10197.
- Junyu Luo, Weizhi Zhang, Ye Yuan, Yusheng Zhao, Junwei Yang, Yiyang Gu, Bohan Wu, Binqi Chen, Ziyue Qiao, Qingqing Long, Rongcheng Tu, Xiao Luo, Wei Ju, Zhiping Xiao, Yifan Wang, Meng Xiao, Chenwu Liu, Jingyang Yuan, Shichang Zhang, and 7 others. 2025. [Large language model agent: A survey on methodology, applications and challenges](#). *CoRR*, abs/2503.21460.
- Yougang Lyu, Xi Zhang, Xinhao Yi, Yuyue Zhao, Shuyu Guo, Wenxiang Hu, Jan Piotrowski, Jakub Kaliski, Jacopo Urbani, Zaiqiao Meng, Lun Zhou, and Xiaohui Yan. 2026. [EvoScientist: Towards multi-agent evolving AI scientists for end-to-end scientific discovery](#). *CoRR*, abs/2603.08127.
- Chengdong Ma, Ziran Yang, Hai Ci, Jun Gao, Minquan Gao, Xuehai Pan, and Yaodong Yang. 2024a. [Evolving diverse red-team language models in multi-round multi-agent games](#). *Preprint*, arXiv:2310.00322.
- Hao Ma, Tianyi Hu, Zhiqiang Pu, Boyin Liu, Xiaolin Ai, Yanyan Liang, and Min Chen. 2024b. [Coevolving with the other you: Fine-tuning LLM with sequential cooperative multi-agent reinforcement learning](#). In *Advances in Neural Information Processing Systems 37: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*.
- Yecheng Jason Ma, William Liang, Guanzhi Wang, De-An Huang, Osbert Bastani, Dinesh Jayaraman, Yuke Zhu, Linxi Fan, and Anima Anandkumar. 2024c. [Eureka: Human-level reward design via coding large language models](#). In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net.
- Rahul Markasserithodi, Aditya Joshi, Yuekang Li, Ishmanbir Singh, Chris Yoo, and Alan Niu. 2026. [CHASE: adversarial red-blue teaming for improving LLM safety using reinforcement learning](#). *CoRR*, abs/2606.05523.
- Tambet Matiisen, Avital Oliver, Taco Cohen, and John Schulman. 2020. [Teacher-student curriculum learning](#). *IEEE Trans. Neural Networks Learn. Syst.*, 31(9):3732–3740.
- Harry Mead, Bruno Lacerda, Jakob N. Foerster, and Nick Hawes. 2025. [Improving regret approximation for unsupervised dynamic environment generation](#). In *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2025, NeurIPS 2025, San Diego, CA, USA, December 2-7, 2025 / Mexico City, Mexico, November 30 - December 5, 2025*.
- Qianyu Meng, Yanan Wang, Liyi Chen, Yihang Li, Wei Wu, Wenyuan Jiang, Qimeng Wang, Chengqiang Lu, Yan Gao, Yi Wu, and 1 others. 2026. [Agent harness for large language model agents: A survey](#).
- Mike A. Merrill, Alexander Glenn Shaw, Nicholas Carlini, Boxuan Li, Harsh Raj, Ivan Bercovich, Lin Shi, Jeong Yeon Shin, Thomas Walshe, Estefany Kelly Buchanan, Junhong Shen, Guanghao Ye, Haowei Lin, Jason Poulos, Maoyu Wang, Marianna Nezhurina, Jena Jitsev, Di Lu, Orfeas Menis-Mastromichalakis, and 66 others. 2026. [Terminal-bench: Benchmarking agents on hard, realistic tasks in command line interfaces](#). *CoRR*, abs/2601.11868.
- Samuel Miserendino, Michele Wang, Tejal Patwardhan, and Johannes Heidecke. 2025. [Swe-lancer: Can frontier llms earn \\$1 million from real-world freelance software engineering?](#) In *Forty-second International Conference on Machine Learning, ICML 2025, Vancouver, BC, Canada, July 13-19, 2025*, volume 267 of *Proceedings of Machine Learning Research*. PMLR / OpenReview.net.
- Konstantinos Mitsides, Maxence Faldor, and Antoine Cully. 2026. [Dreaming in code for curriculum learning in open-ended worlds](#). *CoRR*, abs/2602.08194.
- Nathan Monette, Alistair Letcher, Michael Beukman, Matthew Thomas Jackson, Alexander Rutherford, Alexander David Goldie, and Jakob N. Foerster. 2025. [An optimisation framework for unsupervised environment design](#). *CoRR*, abs/2505.20659.
- Sumeet Ramesh Motwani, Mikhail Baranchuk, Martin Strohmeier, Vijay Bolina, Philip Torr, Lewis Hammond, and Christian Schröder de Witt. 2024. [Secret collusion among AI agents: Multi-agent decep-](#)

- tion via steganography. In *Advances in Neural Information Processing Systems 37: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*.
- Xinyi Mou, Jingcong Liang, Jiayu Lin, Xinnong Zhang, Xiawei Liu, Shiyue Yang, Rong Ye, Lei Chen, Haoyu Kuang, Xuanjing Huang, and Zhongyu Wei. 2025. [Agentsense: Benchmarking social intelligence of language agents through interactive scenarios](#). In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies, NAACL 2025 - Volume 1: Long Papers, Albuquerque, New Mexico, USA, April 29 - May 4, 2025*, pages 4975–5001. Association for Computational Linguistics.
- Xuying Ning, Katherine Tieu, Dongqi Fu, Tianxin Wei, Zihao Li, Yuanchen Bei, Jiaru Zou, Mengting Ai, Zhining Liu, Ting-Wei Li, Lingjie Chen, Yanjun Zhao, Ke Yang, Bingxuan Li, Cheng Qian, Gaotang Li, Xiao Lin, Zhichen Zeng, Ruizhong Qiu, and 23 others. 2026. [Code as agent harness](#). *CoRR*, abs/2605.18747.
- Shayegan Omidshafiei, Dong-Ki Kim, Miao Liu, Gerald Tesauero, Matthew Riemer, Christopher Amato, Murray Campbell, and Jonathan P. How. 2019. [Learning to teach in cooperative multiagent reinforcement learning](#). In *The Thirty-Third AAAI Conference on Artificial Intelligence, AAAI 2019, The Thirty-First Innovative Applications of Artificial Intelligence Conference, IAAI 2019, The Ninth AAAI Symposium on Educational Advances in Artificial Intelligence, EAAI 2019, Honolulu, Hawaii, USA, January 27 - February 1, 2019*, pages 6128–6136. AAAI Press.
- OpenAI, Ilge Akkaya, Marcin Andrychowicz, Maciek Chociej, Mateusz Litwin, Bob McGrew, Arthur Petron, Alex Paino, Matthias Plappert, Glenn Powell, Raphael Ribas, Jonas Schneider, Nikolas Tezak, Jerry Tworek, Peter Welinder, Lilian Weng, Qiming Yuan, Wojciech Zaremba, and Lei Zhang. 2019. [Solving rubik’s cube with a robot hand](#). *Preprint*, arXiv:1910.07113.
- OpenAI, Matthias Plappert, Raul Sampedro, Tao Xu, Ilge Akkaya, Vineet Kosaraju, Peter Welinder, Ruben D’Sa, Arthur Petron, Henrique Pondé de Oliveira Pinto, Alex Paino, Hyeonwoo Noh, Lilian Weng, Qiming Yuan, Casey Chu, and Wojciech Zaremba. 2021. [Asymmetric self-play for automatic goal discovery in robotic manipulation](#). *CoRR*, abs/2101.04882.
- Shuai Pan, Yixiang Liu, Jiaye Gao, Te Gao, Weiwen Liu, Jianghao Lin, Zhihui Fu, Jun Wang, Weinan Zhang, and Yong Yu. 2026a. [Skillmas: Skill co-evolution with llm-based multi-agent system](#). *CoRR*, abs/2605.09341.
- Teng Pan, Yuchen Yan, Zixuan Wang, Ruiqing Zhang, Guiyang Hou, Wenqi Zhang, Weiming Lu, Jun Xiao, and Yongliang Shen. 2026b. [Coverrl: Breaking the consensus trap in label-free reasoning via generator-verifier co-evolution](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2026, San Diego, California, United States, July 2-7, 2026*, pages 29833–29853. Association for Computational Linguistics.
- Zhenyu Pan, Yiting Zhang, Zhuo Liu, Yolo Yunlong Tang, Zeliang Zhang, Haozheng Luo, Yuwei Han, Jianshu Zhang, Dennis Wu, Hong-Yu Chen, Hao-ran Lu, Haoyang Fang, Manling Li, Chenliang Xu, Philip S. Yu, and Han Liu. 2025. [Advevo-marl: Shaping internalized safety through adversarial co-evolution in multi-agent reinforcement learning](#). *CoRR*, abs/2510.01586.
- German Ignacio Parisi, Ronald Kemker, Jose L. Part, Christopher Kanan, and Stefan Wermter. 2019. [Continual lifelong learning with neural networks: A review](#). *Neural Networks*, 113:54–71.
- Chanwoo Park, Seungju Han, Xingzhi Guo, Asuman E. Ozdaglar, Kaiqing Zhang, and Joo-Kyung Kim. 2025. [Maporl: Multi-agent post-co-training for collaborative large language models with reinforcement learning](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2025, Vienna, Austria, July 27 - August 1, 2025*, pages 30215–30248. Association for Computational Linguistics.
- Jongjin Park, Younggyo Seo, Jinwoo Shin, Honglak Lee, Pieter Abbeel, and Kimin Lee. 2022. [Surf: Semi-supervised reward learning with data augmentation for feedback-efficient preference-based reinforcement learning](#). *Preprint*, arXiv:2203.10050.
- Jack Parker-Holder, Minqi Jiang, Michael Dennis, Mikayel Samvelyan, Jakob N. Foerster, Edward Grefenstette, and Tim Rocktäschel. 2022. [Evolving curricula with regret-based environment design](#). In *International Conference on Machine Learning, ICML 2022, 17-23 July 2022, Baltimore, Maryland, USA*, volume 162 of *Proceedings of Machine Learning Research*, pages 17473–17498. PMLR.
- Lerrel Pinto, James Davidson, Rahul Sukthankar, and Abhinav Gupta. 2017. [Robust adversarial reinforcement learning](#). In *Proceedings of the 34th International Conference on Machine Learning, ICML 2017, Sydney, NSW, Australia, 6-11 August 2017*, volume 70 of *Proceedings of Machine Learning Research*, pages 2817–2826. PMLR.
- Aske Plaat, Max J. van Duijn, Niki van Stein, Mike Preuss, Peter van der Putten, and Kees Joost Batenburg. 2025. [Agentic large language models, a survey](#). *J. Artif. Intell. Res.*, 84.
- Rémy Portelas, Cédric Colas, Katja Hofmann, and Pierre-Yves Oudeyer. 2019. [Teacher algorithms for curriculum learning of deep RL in continuously parameterized environments](#). In *3rd Annual Conference*

- on Robot Learning, CoRL 2019, Osaka, Japan, October 30 - November 1, 2019, *Proceedings*, volume 100 of *Proceedings of Machine Learning Research*, pages 835–853. PMLR.
- Zehan Qi, Xiao Liu, Iat Long Iong, Hanyu Lai, Xueqiao Sun, Jiadai Sun, Xinyue Yang, Yu Yang, Shuntian Yao, Wei Xu, Jie Tang, and Yuxiao Dong. 2025. [WebRL: Training LLM web agents via self-evolving online curriculum reinforcement learning](#). In *The Thirteenth International Conference on Learning Representations, ICLR 2025, Singapore, April 24-28, 2025*. OpenReview.net.
- Ao Qu, Han Zheng, Zijian Zhou, Yihao Yan, Yihong Tang, Shao Yong Ong, Fenglu Hong, Kaichen Zhou, Chonghe Jiang, Minwei Kong, Jiacheng Zhu, Xuan Jiang, Sirui Li, Cathy Wu, Bryan Kian Hsiang Low, Jinhua Zhao, and Paul Pu Liang. 2026. [CORAL: towards autonomous multi-agent evolution for open-ended discovery](#). *CoRR*, abs/2604.01658.
- Ben Rank, Hardik Bhatnagar, Ameya Prabhu, Shira Eisenberg, Karina Nguyen, Matthias Bethge, and Maksym Andriushchenko. 2026. [Posttrainbench: Can llm agents automate llm post-training?](#) *Preprint*, arXiv:2603.08640.
- Zhe Ren, Yimeng Chen, Dandan Guo, Guowei Rong, Tonghui Li, R. B. Xiong, Qingfeng Lan, Wenyi Wang, Li Nanbo, Yibo Yang, Mingchen Zhuge, and Jürgen Schmidhuber. 2026. [Self-improvements in modern agentic systems: A survey](#). *Preprint*, arXiv:2607.13104.
- Mikayel Samvelyan, Akbir Khan, Michael Dennis, Minqi Jiang, Jack Parker-Holder, Jakob Nicolaus Foerster, Roberta Raileanu, and Tim Rocktäschel. 2023. [MAESTRO: open-ended environment design for multi-agent reinforcement learning](#). In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net.
- Jürgen Schmidhuber. 2007. [Gödel machines: Fully self-referential optimal universal self-improvers](#). In Ben Goertzel and Cassio Pennachin, editors, *Artificial General Intelligence*, Cognitive Technologies, pages 199–226. Springer.
- David Silver, Julian Schrittwieser, Karen Simonyan, Ioannis Antonoglou, Aja Huang, Arthur Guez, Thomas Hubert, Lucas Baker, Matthew Lai, Adrian Bolton, Yutian Chen, Timothy P. Lillicrap, Fan Hui, Laurent Sifre, George van den Driessche, Thore Graepel, and Demis Hassabis. 2017. [Mastering the game of go without human knowledge](#). *Nat.*, 550(7676):354–359.
- Jiayang Song, Zhehua Zhou, Jiawei Liu, Chunrong Fang, Zhan Shu, and Lei Ma. 2023. [Self-refined large language model as automated reward function designer for deep reinforcement learning in robotics](#). *CoRR*, abs/2309.06687.
- Kenneth O. Stanley, Joel Lehman, and Lisa Soros. 2017. [Open-endedness: The last grand challenge you’ve never heard of](#). *O’Reilly Radar*.
- Shengjie Sun, Runze Liu, Jiafei Lyu, Jingwen Yang, Liangpeng Zhang, and Xiu Li. 2025a. [A large language model-driven reward design framework via dynamic feedback for reinforcement learning](#). *Knowl. Based Syst.*, 326:114065.
- Wangtao Sun, Xiang Cheng, Jialin Fan, Yao Xu, Xing Yu, Shizhu He, Jun Zhao, and Kang Liu. 2025b. [Towards agentic self-learning llms in search environment](#). *Preprint*, arXiv:2510.14253.
- Zeyi Sun, Ziyu Liu, Yuhang Zang, Yuhang Cao, Xiaoyi Dong, Tong Wu, Dahua Lin, and Jiaqi Wang. 2025c. [Seagent: Self-evolving computer use agent with autonomous learning from experience](#). *CoRR*, abs/2508.04700.
- Mirac Suzgun, Mert Yükeşgönül, Federico Bianchi, Dan Jurafsky, and James Zou. 2026. [Dynamic cheat-sheet: Test-time learning with adaptive memory](#). In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics, EACL 2026 - Volume 1: Long Papers, Rabat, Morocco, March 24-29, 2026*, pages 7080–7106. Association for Computational Linguistics.
- Alkis Sygkounas, Rishi Hazra, Andreas Persson, Pedro Zuidberg Dos Martires, and Amy Loutfi. 2026. [Covolve: Adversarial co-evolution of large-language-model-generated policies and environments via two-player zero-sum game](#). In *Proceedings of the Genetic and Evolutionary Computation Conference, GECCO 2026, Centro Internacional de Convenciones CIC-ANDE, San Jose, Costa Rica, July 13-17, 2026*, pages 338–347. ACM.
- Shuhan Tan, Kashyap Chitta, Yuxiao Chen, Ran Tian, Yurong You, Yan Wang, Wenjie Luo, Yulong Cao, Philipp Krähenbühl, Marco Pavone, and Boris Ivanovic. 2025. [Latent chain-of-thought world modeling for end-to-end driving](#). *CoRR*, abs/2512.10226.
- Zhewen Tan, Wenhan Yu, Jianfeng Si, Tongxin Liu, Kaiqi Guan, Huiyan Jin, Jiawen Tao, Xiaokun Yuan, Xiangzheng Zhang, Duohe Ma, Tong Yang, and Lin Sun. 2026. [Triplay-rl: Tri-role self-play reinforcement learning for LLM safety alignment](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2026, San Diego, California, United States, July 2-7, 2026*, pages 26415–26429. Association for Computational Linguistics.
- Open Ended Learning Team, Adam Stooke, Anuj Mahajan, Catarina Barros, Charlie Deck, Jakob Bauer, Jakub Sygnowski, Maja Trebacz, Max Jaderberg, Michaël Mathieu, Nat McAleese, Nathalie Bradley-Schmieg, Nathaniel Wong, Nicolas Porcel, Roberta Raileanu, Steph Hughes-Fitt, Valentin Dalibard, and Wojciech Marian Czarnecki. 2021. [Open-ended learning leads to generally capable agents](#). *CoRR*, abs/2107.12808.

- Kunvar Thaman. 2026. [Reward hacking benchmark: Measuring exploits in LLM agents with tool use](#). *CoRR*, abs/2605.02964.
- Zihang Tian, Jingsen Zhang, Rui Li, Xiaohe Bo, Yuanzi Li, and Xu Chen. 2026. [ARCO: adaptive rubric with co-evolution for multi-step llm-based agents](#). *CoRR*, abs/2606.21262.
- Harsh Trivedi, Tushar Khot, Mareike Hartmann, Ruskin Manku, Vinty Dong, Edward Li, Shashank Gupta, Ashish Sabharwal, and Niranjan Balasubramanian. 2024. [Appworld: A controllable world of apps and people for benchmarking interactive coding agents](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, ACL 2024, Bangkok, Thailand, August 11-16, 2024, pages 16022–16076. Association for Computational Linguistics.
- Mycal Tucker, Roger Levy, Julie A. Shah, and Noga Zaslavsky. 2022. [Trading off utility, informativeness, and complexity in emergent communication](#). In *Advances in Neural Information Processing Systems 35: Annual Conference on Neural Information Processing Systems 2022, NeurIPS 2022, New Orleans, LA, USA, November 28 - December 9, 2022*.
- L. Van Valen. 2014. A new evolutionary law (1973). *Foundations of Macroecology*; University of Chicago Press: Chicago, IL, USA, pages 284–314.
- Oriol Vinyals, Igor Babuschkin, Wojciech M. Czarnecki, Michaël Mathieu, Andrew Dudzik, Junyoung Chung, David H. Choi, Richard Powell, Timo Ewalds, Petko Georgiev, Junhyuk Oh, Dan Horgan, Manuel Kroiss, Ivo Danihelka, Aja Huang, Laurent Sifre, Trevor Cai, John P. Agapiou, Max Jaderberg, and 23 others. 2019. [Grandmaster level in starcraft II using multi-agent reinforcement learning](#). *Nat.*, 575(7782):350–354.
- Hao Wang, Yanting Wang, Hao Li, Rui Li, and Lei Sha. 2026a. [Be your own red teamer: Safety alignment via self-play and reflective experience replay](#). In *Findings of the Association for Computational Linguistics, ACL 2026, San Diego, California, United States, July 2-7, 2026*, pages 18700–18716. Association for Computational Linguistics.
- Rui Wang, Joel Lehman, Jeff Clune, and Kenneth O. Stanley. 2019. [Paired open-ended trailblazer \(POET\): endlessly generating increasingly complex and diverse learning environments and their solutions](#). *CoRR*, abs/1901.01753.
- Shijie Wang, Pengfei Li, Yikun Fu, Kaifeng Liu, Fangyuan Li, Yang Liu, Xiaowei Sun, Zonglin Li, Siyao Zhao, Jian Zhao, Kai Tian, Dong Li, Junqi Gao, Yutong Zhang, Yiqun Chen, Yuqiang Li, Zoe Li, Weinan Zhang, Peng Ye, and 5 others. 2026b. [Marti-mars²: Scaling multi-agent self-search via reinforcement learning for code generation](#). *CoRR*, abs/2602.07848.
- Xucong Wang, Ziyu Ma, Shidong Yang, Tongwen Huang, Pengkun Wang, Yong Wang, and Xiangxiang Chu. 2026c. [Role-agent: Bootstrapping LLM agents via dual-role evolution](#). *CoRR*, abs/2606.10917.
- Yinjie Wang, Tianbao Xie, Ke Shen, Mengdi Wang, and Ling Yang. 2026d. [Rlanything: Forge environment, policy, and reward model in completely dynamic RL system](#). *CoRR*, abs/2602.02488.
- Yinjie Wang, Ling Yang, Ye Tian, Ke Shen, and Mengdi Wang. 2025a. [Co-evolving LLM coder and unit tester via reinforcement learning](#). In *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2025, NeurIPS 2025, San Diego, CA, USA, December 2-7, 2025 / Mexico City, Mexico, November 30 - December 5, 2025*.
- Ziyan Wang, Zhicheng Zhang, Fei Fang, and Yali Du. 2025b. [M3HF: multi-agent reinforcement learning from multi-phase human feedback of mixed quality](#). *CoRR*, abs/2503.02077.
- Ziyi Wang, Yuxuan Lu, Yimeng Zhang, Qun Liu, Chen Luo, Jiri Gesi, Hanqing Lu, Yisi Sang, Manling Li, Jing Huang, and Dakuo Wang. 2026e. [SENTINEL: failure-driven reinforcement learning for training tool-using language model agents](#). *CoRR*, abs/2606.12908.
- Zizhao Wang, Dingcheng Li, Vaishakh Keshava, Phillip Wallis, Ananth Balashankar, Peter Stone, and Lukas Rutishauser. 2025c. [Adversarial reinforcement learning for large language model agent safety](#). *CoRR*, abs/2510.05442.
- Jason Wei, Zhiqing Sun, Spencer Papay, Scott McKinney, Jeffrey Han, Isa Fulford, Hyung Won Chung, Alex Tachard Passos, William Fedus, and Amelia Glaese. 2025. [Browsecomp: A simple yet challenging benchmark for browsing agents](#). *CoRR*, abs/2504.12516.
- Xiaoyu Wen, Zhida He, Han Qi, Ziyu Wan, Zhongtian Ma, Ying Wen, Tianhang Zheng, Xingcheng Xu, Chaochao Lu, and Qiaosheng Zhang. 2026. [MAGIC: A co-evolving attacker-defender adversarial game for robust LLM safety](#). *CoRR*, abs/2602.01539.
- Zhaotian Weng, Antonis Antoniadis, Deepak Nathani, Zhen Zhang, Xiao Pu, and Xin Eric Wang. 2026. [Group-evolving agents: Open-ended self-improvement via experience sharing](#). *CoRR*, abs/2602.04837.
- Jizhou Wu, Jianye Hao, Tianpei Yang, Xiaotian Hao, Yan Zheng, Weixun Wang, and Matthew E. Taylor. 2024. [PORTAL: automatic curricula generation for multiagent reinforcement learning](#). In *Thirty-Eighth AAAI Conference on Artificial Intelligence, AAAI 2024, Thirty-Sixth Conference on Innovative Applications of Artificial Intelligence, IAAI 2024, Fourteenth Symposium on Educational Advances in Artificial Intelligence, EAAI 2024, February 20-27, 2024, Vancouver, Canada*, pages 15934–15942. AAAI Press.

- Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Shaokun Zhang, Erkang Zhu, Beibin Li, Li Jiang, Xiaoyun Zhang, and Chi Wang. 2023. [Autogen: Enabling next-gen LLM applications via multi-agent conversation framework](#). *CoRR*, abs/2308.08155.
- Rong Wu, Xiaoman Wang, Jianbiao Mei, Pinlong Cai, Daocheng Fu, Cheng Yang, Licheng Wen, Xuemeng Yang, Yufan Shen, Yuxin Wang, and Botian Shi. 2025. [Evolver: Self-evolving LLM agents through an experience-driven lifecycle](#). *CoRR*, abs/2510.16079.
- Peng Xia, Kaide Zeng, Jiaqi Liu, Can Qin, Fang Wu, Yiyang Zhou, Caiming Xiong, and Huaxiu Yao. 2025. [Agent0: Unleashing self-evolving agents from zero data via tool-integrated reasoning](#). *CoRR*, abs/2511.16043.
- Zhishang Xiang, Chengyi Yang, Zerui Chen, Zhimin Wei, Yunbo Tang, Zongpei Teng, Zexi Peng, Zongxia Li, Chengsong Huang, Yicheng He, Chang Yang, Xinrun Wang, Xiao Huang, Qinggang Zhang, and Jin-song Su. 2026. [A systematic survey of self-evolving agents: From model-centric to environment-driven co-evolution](#). *TechRxiv*, 2026(0227).
- Chuan Xiao, Zhengbo Jiao, Shaobo Wang, Wei Wang, Bing Zhao, Hu Wei, Linfeng Zhang, and Lin Qu. 2026. [Socratic-swe: Self-evolving coding agents via trace-derived agent skills](#). *CoRR*, abs/2606.07412.
- Tianbao Xie, Danyang Zhang, Jixuan Chen, Xiaochuan Li, Siheng Zhao, Ruisheng Cao, Toh Jing Hua, Zhoujun Cheng, Dongchan Shin, Fangyu Lei, Yitao Liu, Yiheng Xu, Shuyan Zhou, Silvio Savarese, Caiming Xiong, Victor Zhong, and Tao Yu. 2024. [Osworld: Benchmarking multimodal agents for open-ended tasks in real computer environments](#). In *Advances in Neural Information Processing Systems 37: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*.
- Zhongwei Xie, Xiyu Ren, Tianshi Zheng, Jiaxin Bai, Wei Fan, Baixuan Xu, Haoran Li, Huihao Jing, and Yangqiu Song. 2026. A survey on ai agent harness. *ResearchGate Preprint*. DOI, 10.
- Manjie Xu, Xinyi Yang, Jiayu Zhan, Wei Liang, Chi Zhang, and Yixin Zhu. 2025a. [Heterogeneous adversarial play in interactive environments](#). In *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2025, NeurIPS 2025, San Diego, CA, USA, December 2-7, 2025 / Mexico City, Mexico, November 30 - December 5, 2025*.
- Weikai Xu, Chengrui Huang, Shen Gao, and Shuo Shang. 2025b. [Llm-based agents for tool learning: A survey](#). *Data Sci. Eng.*, 10(4):533–563.
- Taofeng Xue, Chong Peng, Mianqiu Huang, Linsen Guo, Tiancheng Han, Haozhe Wang, Jianing Wang, Xiaocheng Zhang, Xin Yang, Dengchang Zhao, Jinrui Ding, Xiandi Ma, Yuchen Xie, Peng Pei, Xunliang Cai, and Xipeng Qiu. 2026a. [Evocua: Evolving computer use agents via learning from scalable synthetic experience](#). *CoRR*, abs/2601.15876.
- Tianci Xue, Zeyi Liao, Tianneng Shi, Zilu Wang, Kai Zhang, Dawn Song, Yu Su, and Huan Sun. 2026b. [Autonomous continual learning of computer-use agents for environment adaptation](#). *CoRR*, abs/2602.10356.
- Xiangyuan Xue, Yifan Zhou, Guibin Zhang, Zaibin Zhang, Yijiang Li, Chen Zhang, Zhenfei Yin, Philip Torr, Wanli Ouyang, and Lei Bai. 2025. [Comas: Co-evolving multi-agent systems via interaction rewards](#). *CoRR*, abs/2510.08529.
- Jiachen Yang, Ang Li, Mehrdad Farajtabar, Peter Sunehag, Edward Hughes, and Hongyuan Zha. 2020. [Learning to incentivize other learning agents](#). In *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual*.
- Shidong Yang, Ziyu Ma, Tongwen Huang, Yiming Hu, Yong Wang, and Xiangxiang Chu. 2026. [Coevolve: Training LLM agents via agent-data mutual evolution](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2026, San Diego, California, United States, July 2-7, 2026*, pages 23015–23036. Association for Computational Linguistics.
- Yaodong Yang, Rui Luo, Minne Li, Ming Zhou, Weinan Zhang, and Jun Wang. 2018. [Mean field multi-agent reinforcement learning](#). In *Proceedings of the 35th International Conference on Machine Learning, ICML 2018, Stockholm, Sweden, July 10-15, 2018*, volume 80 of *Proceedings of Machine Learning Research*, pages 5567–5576. PMLR.
- Changwei Yao, Xinzi Liu, Chen Li, and Marios Savvides. 2026a. [Reward evolution with graph-of-thoughts: A bi-level language model framework for reinforcement learning](#). *Preprint*, arXiv:2509.16136.
- Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. 2024. [\$\tau\$ -bench: A benchmark for tool-agent-user interaction in real-world domains](#). *CoRR*, abs/2406.12045.
- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik R. Narasimhan, and Yuan Cao. 2023. [React: Synergizing reasoning and acting in language models](#). In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net.
- Yilun Yao, Xinyu Tan, Chao-Hsuan Liu, Yaoming Li, Zhengyang Wang, Wenhan Yu, Zhewen Tan, Yuxuan Tian, Guangxiang Zhao, Lin Sun, Xiangzheng Zhang, and Tong Yang. 2026b. [Harness-bench: Measuring harness effects across models in realistic agent workflows](#). *CoRR*, abs/2605.27922.

- Teresa Yeo, Dulaj Weerakoon, Dulanga Weerakoon, and Archan Misra. 2026. [Towards adaptive environment generation for training embodied agents](#). *CoRR*, abs/2602.06366.
- Xunjian Yin, Xinyi Wang, Liangming Pan, Li Lin, Xiaojun Wan, and William Yang Wang. 2025. [Gödel agent: A self-referential agent framework for recursively self-improvement](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2025, Vienna, Austria, July 27 - August 1, 2025*, pages 27890–27913. Association for Computational Linguistics.
- Zhenrui Yue, Kartikeya Upasani, Xianjun Yang, Suyu Ge, Shaoliang Nie, Yuning Mao, Zhe Liu, and Dong Wang. 2026. [Dr. zero: Self-evolving search agents without training data](#). *CoRR*, abs/2601.07055.
- Taeyoung Yun, Pierre-Luc St-Charles, Jinkyoo Park, Yoshua Bengio, and Minsu Kim. 2025. [Active attacks: Red-teaming llms via adaptive environments](#). *CoRR*, abs/2509.21947.
- Abhay Zala, Jaemin Cho, Han Lin, Jaehong Yoon, and Mohit Bansal. 2024. [Envgen: Generating and adapting environments via llms for training embodied agents](#). *CoRR*, abs/2403.12014.
- Kaiwen Zha, Zhengqi Gao, Maohao Shen, Zhang-Wei Hong, Duane S. Boning, and Dina Katabi. 2025. [RL tango: Reinforcing generator and verifier together for language reasoning](#). In *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2025, NeurIPS 2025, San Diego, CA, USA, December 2-7, 2025 / Mexico City, Mexico, November 30 - December 5, 2025*.
- Yunpeng Zhai, Shuchang Tao, Cheng Chen, Anni Zou, Ziqian Chen, Qingxu Fu, Shinji Mai, Li Yu, Jiaji Deng, Zouying Cao, Zhaoyang Liu, Bolin Ding, and Jingren Zhou. 2025a. [Agentevolver: Towards efficient self-evolving agent system](#). *CoRR*, abs/2511.10395.
- Yunpeng Zhai, Shuchang Tao, Cheng Chen, Anni Zou, Ziqian Chen, Qingxu Fu, Shinji Mai, Li Yu, Jiaji Deng, Zouying Cao, Zhaoyang Liu, Bolin Ding, and Jingren Zhou. 2025b. [Agentevolver: Towards efficient self-evolving agent system](#). *CoRR*, abs/2511.10395.
- Guibin Zhang, Haotian Ren, Chong Zhan, Zhenhong Zhou, Junhao Wang, He Zhu, Wangchunshu Zhou, and Shuicheng Yan. 2025a. [Memevolve: Meta-evolution of agent memory systems](#). *CoRR*, abs/2512.18746.
- Hanrong Zhang, Shicheng Fan, Henry Peng Zou, Yankai Chen, Zhenting Wang, Jiayu Zhou, Chengze Li, Wei-Chieh Huang, Yifei Yao, Kening Zheng, Xue Liu, Xiaoxiao Li, and Philip S. Yu. 2026a. [Coevoskills: Self-evolving agent skills via co-evolutionary verification](#). *Preprint*, arXiv:2604.01687.
- Jenny Zhang, Bingchen Zhao, Wannan Yang, Jakob N. Foerster, Jeff Clune, Minqi Jiang, Sam Devlin, and Tatiana Shavrina. 2026b. [Hyperagents](#). *CoRR*, abs/2603.19461.
- Jingyu Zhang, Haozhu Wang, Eric Michael Smith, Sid Wang, Amr Sharaf, Mahesh Pasupuleti, Benjamin Van Durme, Daniel Khashabi, Jason Weston, and Hongyuan Zhan. 2025b. [The alignment waltz: Jointly training agents to collaborate for safety](#). *CoRR*, abs/2510.08240.
- Ruize Zhang, Zelai Xu, Chengdong Ma, Chao Yu, Weiwei Tu, Shiyu Huang, Deheng Ye, Wenbo Ding, Yaodong Yang, and Yu Wang. 2024. [A survey on self-play methods in reinforcement learning](#). *CoRR*, abs/2408.01072.
- Yaolun Zhang, Yujie Zhao, Nan Wang, Yiran Wu, Jiayu Chang, Yizhao Chen, Qingyun Wu, Jishen Zhao, and Huazheng Wang. 2026c. [Metaagent-x : Breaking the ceiling of automatic multi-agent systems via end-to-end reinforcement learning](#). *Preprint*, arXiv:2605.14212.
- Wanjia Zhao, Mert Yüsekşgönül, Shirley Wu, and James Y. Zou. 2025. [Sirius: Self-improving multi-agent systems via bootstrapped reasoning](#). In *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2025, NeurIPS 2025, San Diego, CA, USA, December 2-7, 2025 / Mexico City, Mexico, November 30 - December 5, 2025*.
- Yujie Zhao, Lanxiang Hu, Yang Wang, Minmin Hou, Hao Zhang, Ke Ding, and Jishen Zhao. 2026. [Stronger-mas: Multi-agent reinforcement learning for collaborative llms](#). *Preprint*, arXiv:2510.11062.
- Stephan Zheng, Alexander Trott, Sunil Srinivasa, Nikhil Naik, Melvin Gruesbeck, David C. Parkes, and Richard Socher. 2020. [The AI economist: Improving equality and productivity with ai-driven tax policies](#). *CoRR*, abs/2004.13332.
- Shuyan Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou, Robert Lo, Abishek Sridhar, Xianyi Cheng, Tianyue Ou, Yonatan Bisk, Daniel Fried, Uri Alon, and Graham Neubig. 2024a. [Webarena: A realistic web environment for building autonomous agents](#). In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net.
- Xuhui Zhou, Hao Zhu, Leena Mathur, Ruohong Zhang, Haofei Yu, Zhengyang Qi, Louis-Philippe Morency, Yonatan Bisk, Daniel Fried, Graham Neubig, and Maarten Sap. 2024b. [SOTOPIA: interactive evaluation for social intelligence in language agents](#). In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net.
- Yifei Zhou, Sergey Levine, Jason E. Weston, Xian Li, and Sainbayar Sukhbaatar. 2025. [Self-challenging](#)

language model agents. In *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2025, NeurIPS 2025, San Diego, CA, USA, December 2-7, 2025 / Mexico City, Mexico, November 30 - December 5, 2025*.

Yingli Zhou, Wang Shu, Yaodong Su, Wenchuan Du, Yixiang Fang, and Xuemin Lin. 2026. [A comprehensive survey on agent skills: Taxonomy, techniques, and applications](#). *CoRR*, abs/2605.07358.

Kunlun Zhu, Hongyi Du, Zhaochen Hong, Xiaocheng Yang, Shuyi Guo, Zhe Wang, Zhenhailong Wang, Cheng Qian, Robert Tang, Heng Ji, and Jiaxuan You. 2025a. [Multiagentbench : Evaluating the collaboration and competition of LLM agents](#). In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, ACL 2025, Vienna, Austria, July 27 - August 1, 2025, pages 8580–8622. Association for Computational Linguistics.

Yiwen Zhu, Jinyi Liu, Pengjie Gu, Yifu Yuan, Zhenxing Ge, Wenya Wei, Zhou Fang, Yujing Hu, and Bo An. 2025b. [Improving reward models with proximal policy exploration for preference-based reinforcement learning](#). In *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2025, NeurIPS 2025, San Diego, CA, USA, December 2-7, 2025 / Mexico City, Mexico, November 30 - December 5, 2025*.

Qing Zong, Yue Guo, Mengxin Yang, Yiwen Guo, and Yangqiu Song. 2026. [Evolvingworld: An open-schema framework for co-evolving role-play agents and world model in interactive literary world](#). *arXiv preprint arXiv:2607.17250*.

附录

区分协同演化与相邻概念

多智能体交互。多个智能体可以在不改变自身的情况下进行通信、协作或竞争（Wu et al., 2023; Guo et al., 2024）。只有当交互对多个智能体产生持续更新，并且它们的变化继续相互塑造时，交互才成为共同进化。

智能体循环与循环工程。智能体循环组织用于完成任务所重复执行的步骤（Xie 等, 2026）。循环工程设计该过程的目标、反馈、验证和停止规则。两者都关注任务如何被执行，而非智能体是否跨任务发生变化。

框架工程。框架工程设计智能体的持久运行时组件，包括其提示、记忆、工具、技能和工作流（Xie 等, 2026）。仅设计框架并不构成进化。只有当模型、框架或两者在多次运行中被更新并保留时，进化才会发生（Liu 等, 2026i）。即使两者都发生变化，在我们的定义中它们仍构成一个进化的智能体。

进化优化。进化优化通过生成、评估和选择来搜索候选提示、程序或智能体设计（Chauhan 等, 2025）。其候选者只需被评估和选择，而协同进化则需要能够响应彼此变化的持久自适应单元。

持续学习。持续学习在新数据或任务到来时反复更新同一个学习器并保留其变化（Parisi 等, 2019）。与进化优化不同，它在一个持续的学习器内累积变化，而不是在独立的候选者之间进行选择。其学习过程可以保持不变，且无需其他组件进行适应。

自我博弈。自我博弈是一种训练方案，其中学习者与自身的副本或早期版本进行交互（Silver 等人, 2017; Zhang 等人, 2024）。即使只有当前学习者发生变化，它也能支持自我进化。相比之下，共同进化需要多个自适应单元，这些单元保留其变化并持续相互塑造。

综述 主要范围 原始分类法 共同进化的覆盖范围			
Meng et al. (2026)	Runtime infrastructure that governs LLM-agent execution	Where the harness sits in the agent stack and whether it is general or domain-specific	No dedicated coverage. The survey is organized around harness components and contains no co-evolution category or section.
Guo et al. (2024)	Design, communication, and collaboration in LLM-based multi-agent systems	Agents–Environment Interface; Agents Profiling; Agents Communication; Agents Capability Acquisition	No dedicated coverage. Section 3.4 lists self-evolution as one strategy for updating agent capabilities, but the survey has no section on co-evolution.
Huang et al. (2025c)	How environments generate tasks, support agent execution, and provide training feedback	Task Generation; Task Execution; Feedback	Mainly discussed as a future direction. Section 6 includes “Co-Evolution via Embedded External Tools,” but co-evolution is not part of the main taxonomy.
Fang (2025a)	Techniques for self-evolving agentic systems and their domain-specific applications	System Inputs; Agent System; Environment; Optimisers	Mentioned as an open challenge. Section 8.1.3 identifies the co-evolution of tools alongside agents as underexplored, but it is not part of the main taxonomy.
Ren et al. (2026)	Self-improvement of the foundation model and its operational scaffold	Foundation Model Improvement; Scaffolding Improvement	Mainly discussed as a future direction. Section 9.2 lists Multi-Agent Cooperative Co-Evolution as one research direction, but it is not part of the main taxonomy.
Xie et al. (2026)	Architecture and evolution of the infrastructure surrounding an agent model	Execution & Orchestration; Context & Trajectory Management; Interaction Surface & Execution Environment; Constraints & Guardrails	Discussed as a cross-cutting design pattern. It covers model–harness co-evolution, but under our definition, changes to both still constitute the evolution of a single agent.
Chen et al. (2026b)	Recursive self-improvement from bounded refinement to autonomous AI research	Deployment-Time Self-Evolution; Training-Time Self-Iteration; Self-Evaluation; Auto Research, crossed with Human-in-the-loop, Human-on-the-loop, and Closed loop	Covered through scattered examples. Co-evolution appears in self-play and evaluator evolution, but it is not part of the main taxonomy.
Gao et al. (2026)	Self-evolving agents, including single-agent component updates and multi-agent methods	What, When, How, and Where to Evolve	Covered across several subsections. Sections 3.4.2, 5.3.2, and 6.1 cover multi-agent and model–agent evolution without a unified co-evolution taxonomy.
Xiang et al. (2026)	The broader field of self-evolving agents, including model-centric and environment-centric single-entity evolution	Model-Centric Self-Evolution; Environment-Centric Self-Evolution; Model–Environment Co-Evolution	Included as one top-level category. Model–Environment Co-Evolution is one of three branches, but the survey does not cover Agent–Agent or Meta Co-Evolution.
Ours	Mutual adaptation among components in agentic systems	Agent–Agent Co-Evolution; Agent–Environment Co-Evolution; Meta Co-Evolution	Co-evolution is the central subject and organizing principle. We cover adaptive agents, environments, and evolution mechanisms.

表 1: 与相邻领域综述的比较。我们保留每篇综述的原始分类法，并报告共同进化在其组织结构中出现的位置。

自我进化。自我进化允许智能体在其任务、反馈规则、对手或其他学习条件保持不变的情况下进行改进（Gao 等人，2026）。共同进化需要另一个自适应单元，该单元的变化会重塑智能体后续的进化，同时智能体也会重塑该单元。这个额外的单元可以由智能体自身创建，但在过程中它必须作为一个独立的部分进行进化。

元进化。元进化改变支配未来进化的机制，包括进化什么、何时进化、如何进化、在哪里进化，以及如何评估变化（Fernando 等人，2024；Chen 等人，2026b）。元层级控制器可能

只要它修改这一机制，就保持不变。元协同演化进一步要求这种修改由低层协同演化系统驱动，并反过来重塑该系统。

迭代式与递归式自我改进。迭代式自我改进反复应用固定的改进程序使系统变得更好。递归式自我改进还改进这一程序本身，因此更新后的系统能在后续轮次中进行更有效的自我改进（Schmidhuber, 2007; Chen et al., 2026b）。两者都可能发生在单一系统内，因此不一定构成协同演化。

开放性。开放性意味着系统持续产生新颖且可学习的结果，而不是停留在固定终点（Stanley et al., 2017; Hughes et al., 2024）。它描述了一种可能的长期结果，而非某种特定的演化形式，协同演化并不自动保证开放性。

B 与相关综述的比较

表 1 使用各综述原有的分类法，将本综述与相邻综述进行比较。现有综述主要围绕多智能体系统、智能体框架、自我演化、递归改进或环境扩展来组织文献。因此，协同演化以零散示例、未来方向或更广泛分类法中的一个分支的形式出现。相比之下，本文以协同演化作为核心纳入标准，并区分其发生在智能体、环境和演化机制之间的位置。

C 跨论文证据图的构建

图 4 汇总了先前工作所报告的结果，而非新实验的结果。由于所纳入的论文使用了不同的模型、基准和评估度量，本文并未直接合并其原始分数。相反，所有比较和聚合均首先在各论文内部进行。

面板 A 和面板 B 中的配对比较。仅当一篇论文在同一骨干网络和评估设置下同时报告了静态或冻结对照与演化对照的结果时，本文才将其纳入。当存在多个基线时，本文选择与完整方法的主要差异仅在于相关对照是否允许演化的受控变体，而非任意或最强的基线。缺少两种条件结果的评估设置被排除。对于面板 A，本文分别计算静态条件和演化条件下配对设置的平均分数。在平均之前对度量进行方向对齐，使得更高的值始终表示更好的性能。因此，两个端点应仅被解释为论文内部的比较；其绝对位置在不同论文之间不可直接比较。面板 B 使用相同的配对结果。其水平坐标为配对评估设置的数量，垂直坐标为启用

对照演化所产生的平均配对改进。气泡面积表示具有正向改进的配对设置的比例，而颜色表示本文分类法中的阶段。任何未配对或仅聚合的结果均不被视为单独的评估设置。

面板 C 中的归一化轨迹。面板 C 由跨连续演化轮次、训练轮次或训练检查点报告的性能轨迹构建而成。当存在明确报告的标签时，数值取自这些标签；否则从已发表的图表中近似数字化。对于每条轨迹，演化进度被归一化到区间 $[0, 1]$ ：

$$\hat{x}_t = \frac{x_t - x_0}{x_T - x_0}.$$

性能表示为最大观测改进的分数：

$$\hat{y}_t = \frac{y_t - y_0}{\max_{\tau} y_{\tau} - y_0}.$$

越低越好的度量在归一化前进行方向对齐。未观察到正向改进的轨迹不纳入分析。每条归一化轨迹被线性插值到统一的进度网格上。当一篇论文报告多个模型或评估度量时，这些轨迹首先在该论文内取平均，使得每篇论文贡献一条等权重的曲线。面板 C 中的实线是论文级轨迹的均值，阴影区域表示跨论文的一个样本标准差。该区间反映的是跨论文的异质性，而非实验不确定性或置信区间。